

## Lecture 13

# Plane Motion of Rigid Bodies: Forces and Accelerations

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MC2103  
Fall 2026





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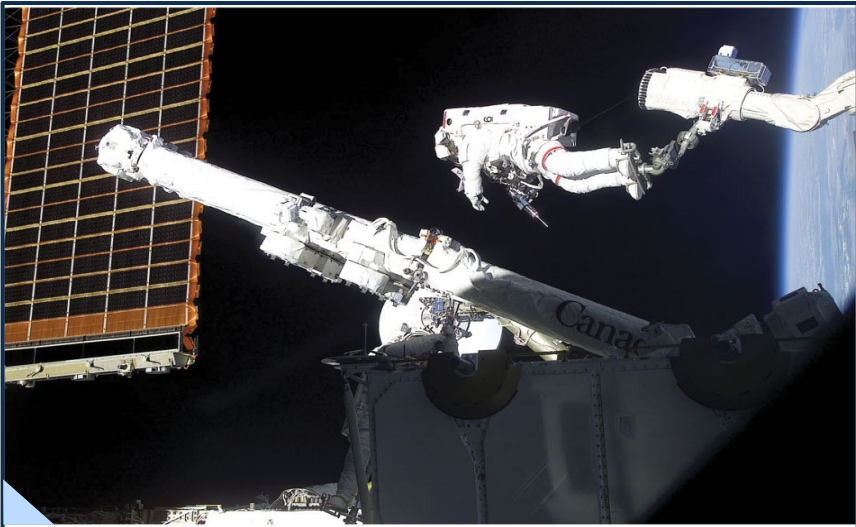
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## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Rigid Body Kinetics

The forces and moments applied to a robotic arm control the **resulting kinematics**, and therefore the end position and forces of the actuator at the end of the robot arm.

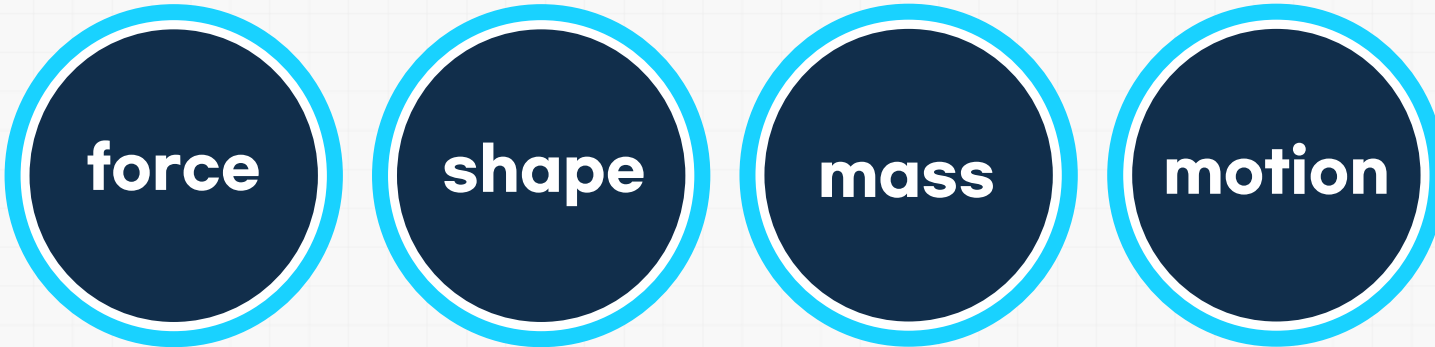


## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Introduction

⚙️ **The forces, the shape and mass of the body, and the motion**

- We will be concerned with the kinetics of rigid bodies, that is relations between **the forces** acting on a rigid body, **the shape and mass of the body**, and **the motion** produced.



force

shape

mass

motion

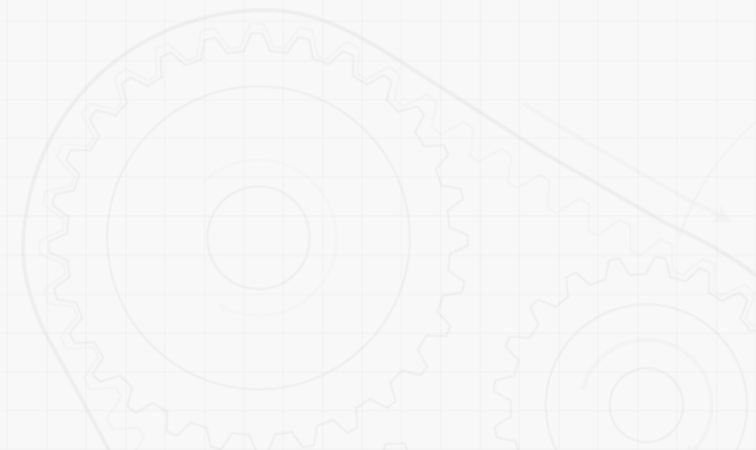


## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Introduction

⚙️ **Results of this chapter will be restricted to**

- Plane motion of rigid bodies
- Rigid bodies consisting of plane slabs or bodies which are symmetrical with respect to the reference plane



## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Introduction

## ⚙ Approach

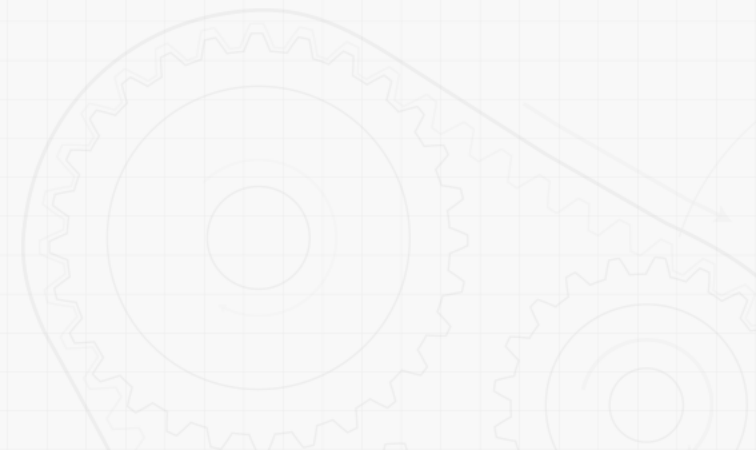
- Consider **rigid bodies** as made of large numbers of particles
- Use the results of Week 8 for **the motion of systems of particles**

Specifically,

$$\sum \vec{F} = m\vec{a}$$

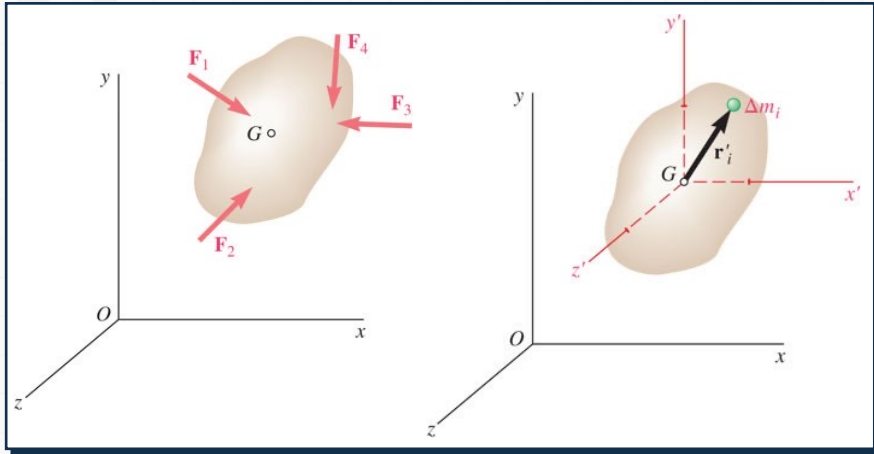
and

$$\sum \vec{M}_G = \dot{\vec{H}}_G$$



## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Equations of Motion for a Rigid Body



- Consider a rigid body acted upon by several external forces.
- Assume that the body is made of a large number of particles.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

## From 3D Space to Planar Motion

⚙️ **Dynamics in 2D space is a special case of 3D dynamics.**

- Angular Momentum of a body  $B$  about  $G$  in 3D space

$$\begin{aligned}\vec{r} &= r_1 \hat{b}_1 + r_2 \hat{b}_2 + r_3 \hat{b}_3 \\ {}^N\vec{\omega}^B &= \omega_1 \hat{b}_1 + \omega_2 \hat{b}_2 + \omega_3 \hat{b}_3\end{aligned}$$

$$\vec{H}_G = \begin{pmatrix} H_1 \\ H_2 \\ H_3 \end{pmatrix}_B = \begin{pmatrix} \int_B (r_2^2 + r_3^2) dm & -\int_B r_1 r_2 dm & -\int_B r_1 r_3 dm \\ -\int_B r_2 r_1 dm & \int_B (r_1^2 + r_3^2) dm & -\int_B r_2 r_3 dm \\ -\int_B r_3 r_1 dm & -\int_B r_3 r_2 dm & \int_B (r_1^2 + r_2^2) dm \end{pmatrix} \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix} = \mathbf{I}_G \boldsymbol{\omega}$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# From 3D Space to Planar Motion

⚙️ **Dynamics in 2D space is a special case of 3D dynamics.**

- Angular Momentum of a body  $B$  about  $G$  in 2D space

$$\begin{aligned}\vec{r} &= r_1 \hat{b}_1 + r_2 \hat{b}_2 + 0 \hat{b}_3 \\ {}^N \vec{\omega}^B &= 0 \hat{b}_1 + 0 \hat{b}_2 + \omega_3 \hat{b}_3\end{aligned}$$

$$\vec{H}_G = \begin{pmatrix} H_1 \\ H_2 \\ H_3 \end{pmatrix}_B = \begin{pmatrix} \int_B r_2^2 dm & -\int_B r_1 r_2 dm & 0 \\ -\int_B r_2 r_1 dm & \int_B r_1^2 dm & 0 \\ 0 & 0 & \int_B (r_1^2 + r_2^2) dm \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ \omega_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \int_B (r_1^2 + r_2^2) dm \omega_3 \end{pmatrix}$$

$$H = \int_B (r_1^2 + r_2^2) dm \omega = I_G \omega$$

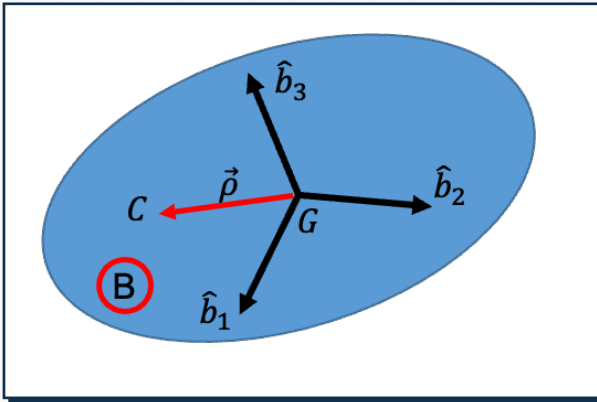
$H, \omega$  are scalar, not a vector.  $I_G$  is a scalar, not a 3x3 matrix

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Parallel Axis Theorem

⚙️ **Dynamics in 2D space is a special case of 3D dynamics.**

- Moment of inertia matrix about  $C$



$$I_C = I_G + m[(\vec{\rho} \cdot \vec{\rho})I_3 - \vec{\rho} \otimes \vec{\rho}]$$

$$= I_G + m \begin{pmatrix} \rho_2^2 + \rho_3^2 & -\rho_1\rho_2 & -\rho_1\rho_3 \\ -\rho_2\rho_1 & \rho_1^2 + \rho_3^2 & -\rho_2\rho_3 \\ -\rho_3\rho_1 & -\rho_3\rho_2 & \rho_1^2 + \rho_2^2 \end{pmatrix}$$

- Where  $\vec{\rho} = \rho_1\hat{b}_1 + \rho_2\hat{b}_2 + \rho_3\hat{b}_3$ ,  $\otimes$  is the outer product,  $I_3$  is the 3-by-3 identity matrix.

$$I_C = I_G + m(\rho_1^2 + \rho_2^2) = I_G + m|\rho|^2$$

$I_G$  and  $I_C$  are a scalar, not a 3x3 matrix

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Summary of Newton-Euler Equations

## ⚙ For translational motion

$$\sum_i \vec{F}_i = {}^N \dot{\vec{L}}, \quad \text{where } \vec{L} = m\vec{v}_G$$



$$\sum_i \vec{F}_i = {}^N \dot{\vec{L}}, \quad \text{where } \vec{L} = m\vec{v}_G$$

No change

## ⚙ For rotational motion about COM, $G$

$$\begin{aligned} \sum_i \vec{M}_{G,i} &= {}^N \dot{\vec{H}}_G, & \text{where } \vec{H}_G &= I_G \omega \\ &= {}^B \dot{\vec{H}}_G + {}^N \vec{\omega}^B \times \vec{H}_G \\ &= I_G \dot{\omega} + \omega \times I_G \omega \end{aligned}$$

$$\begin{aligned} \sum_i M_{G,i} &= {}^N \dot{H}_G, & \text{where } H_G &= I_G \omega \\ &= I_G \dot{\omega} \\ &= I_G \alpha \end{aligned}$$

$M_G, H_G, I_G, \omega, \alpha$  are all scalar.

Also, note that  ${}^N \vec{\omega}^B \times \vec{H}_G = 0$  since both are aligned.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Summary of Newton-Euler Equations

⚙ For rotational motion about arbitrary point,  $C$

$${}^N\dot{\vec{H}}_C = \sum_i \vec{M}_{C,i} + m\vec{v}_G \times \vec{v}_C, \quad \text{where } \vec{H}_C = I_C\omega + \vec{\rho} \times m\vec{v}_C$$

• If  $C$  is a fixed point,

$${}^N\dot{\vec{H}}_C = \sum_i \vec{M}_{C,i}, \quad \text{where } \vec{H}_C = I_C\omega$$

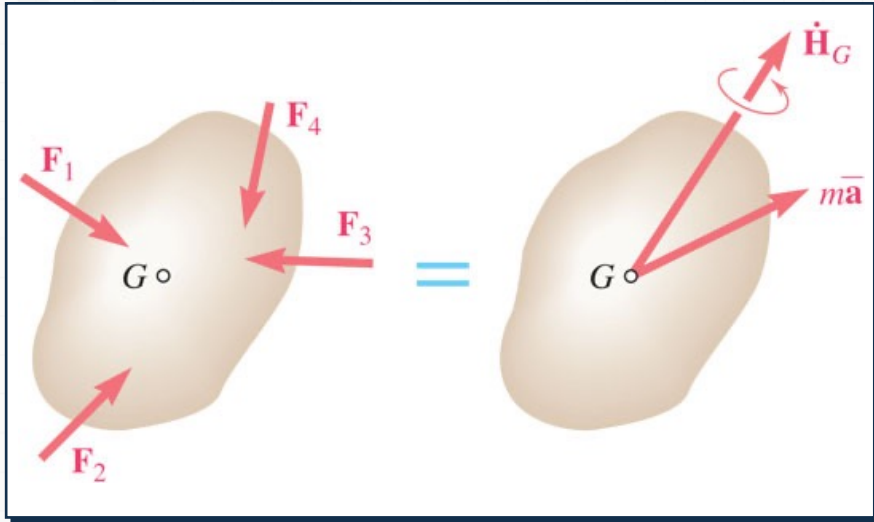
$$\begin{aligned} \sum_i M_{C,i} &= {}^N\dot{H}_C, & \text{where } H_C &= I_C\omega \\ &= I_C\dot{\omega} \\ &= I_C\alpha \end{aligned}$$

$M_C, H_C, I_C, \omega, \alpha$  are all scalar.

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## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

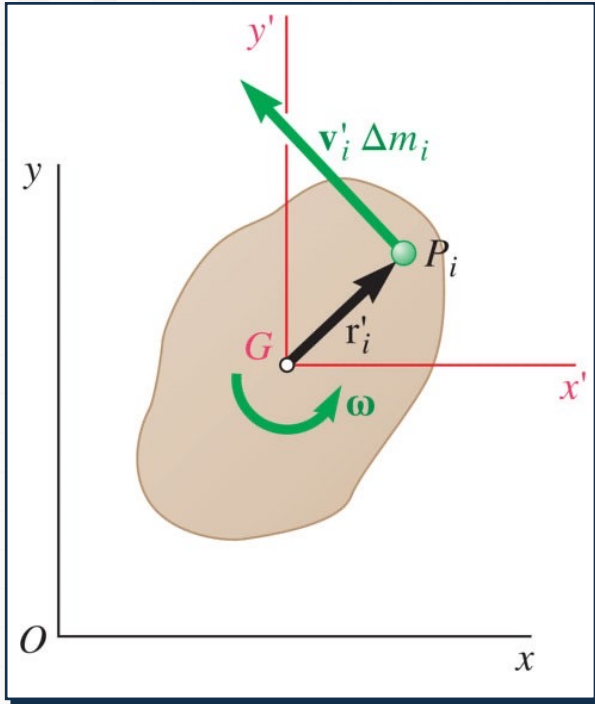
# Equations of Motion for a Rigid Body



- System of external forces is equipollent to the system consisting of  $m\vec{a}$  and  $\dot{H}_G$ .

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

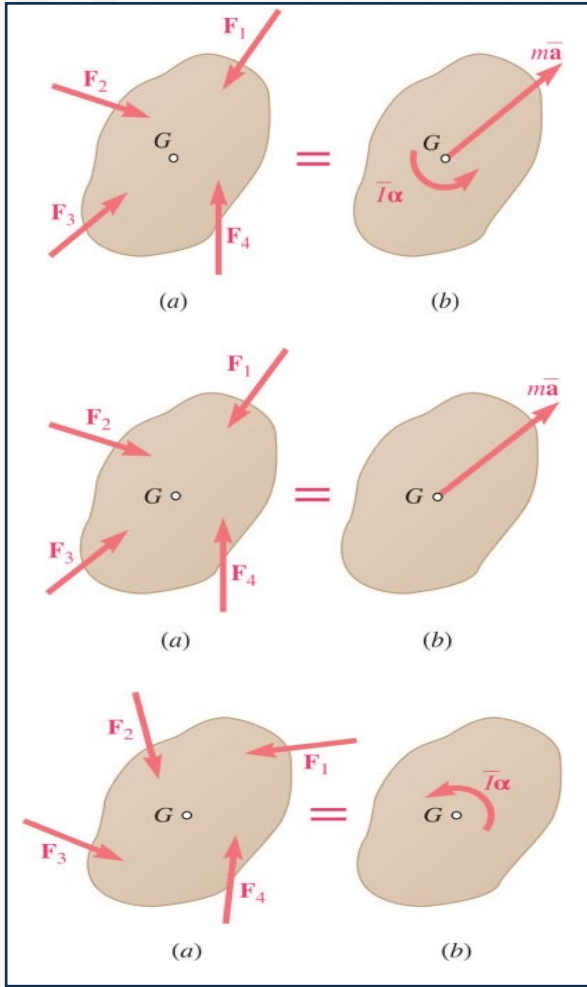
# Angular Momentum of a Rigid Body in Plane Motion



- Results are also valid for **plane motion** of bodies which are symmetrical with respect to the reference plane.
- Results are not valid for **asymmetrical bodies** or **three-dimensional motion**.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Plane Motion of a Rigid Body



- Motion of a rigid body in plane motion is completely defined by **the resultant and moment resultant about G of the external forces.**

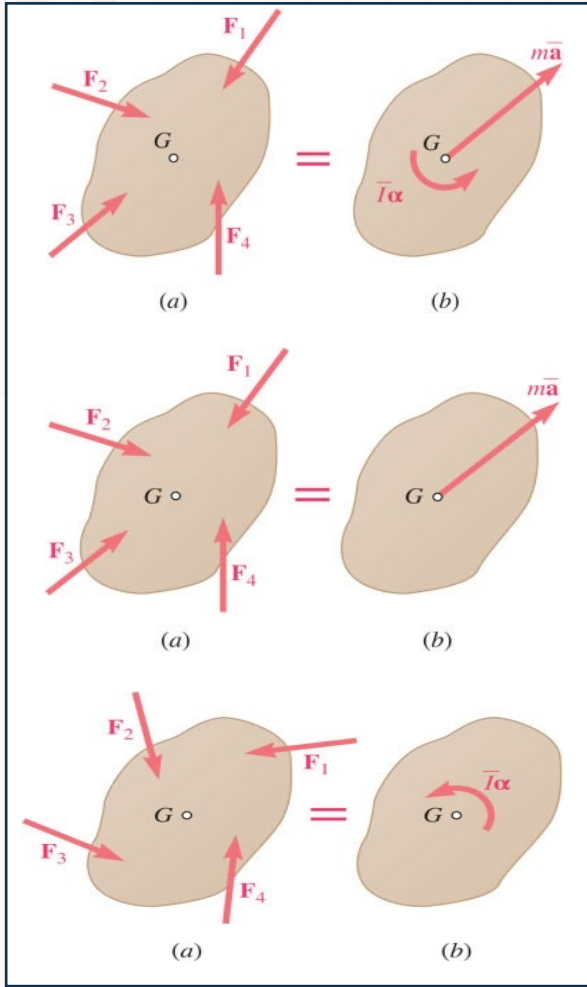
$$\sum F_x = m\bar{a}_x$$

$$\sum F_y = m\bar{a}_y$$

$$\sum M_G = \bar{I}a$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

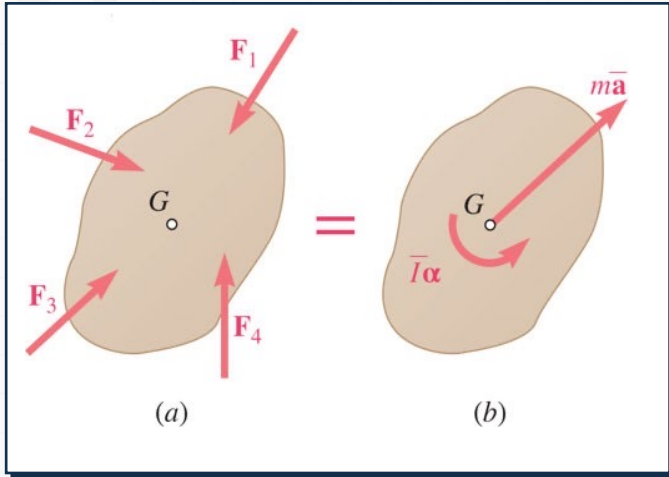
# Plane Motion of a Rigid Body



- The most general motion of a rigid body that is symmetrical with respect to the reference plane can be replaced by the sum of a translation and a centroidal rotation.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

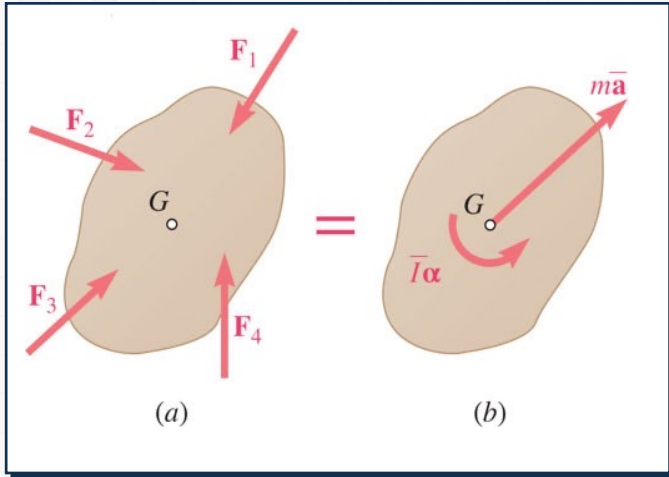
# Problems Involving the Motion of a Rigid Body



- The fundamental relation between the forces acting on a rigid body in plane motion and the acceleration of its mass center and the angular acceleration of the body is illustrated in a **free-body-diagram equation**.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Problems Involving the Motion of a Rigid Body



• This technique may be applied as follows.

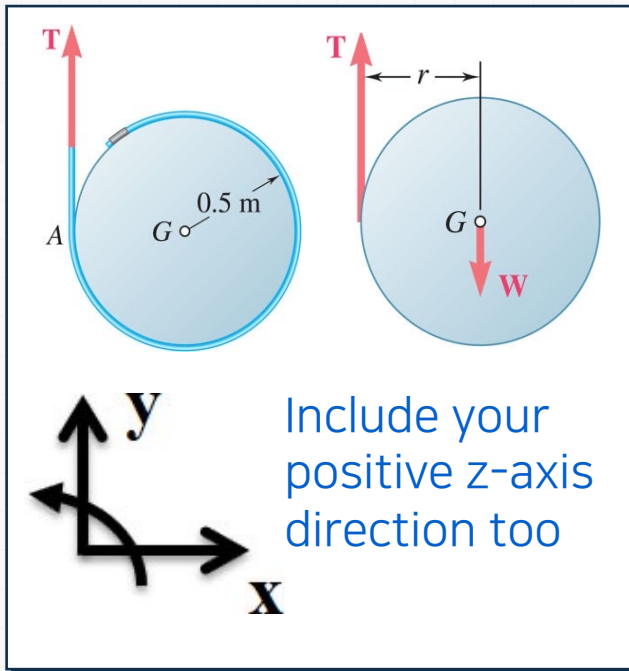
- Problems involving plane motion of connected rigid bodies by drawing a free-body-diagram equation for each body and solving the corresponding equations of motion simultaneously.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Free Body Diagrams and Kinetic Diagrams

## ⚙ The free body diagram

- The free body diagram is the same as you have done in statics and in Week 4; we will add the kinetic diagram in our dynamic analysis.

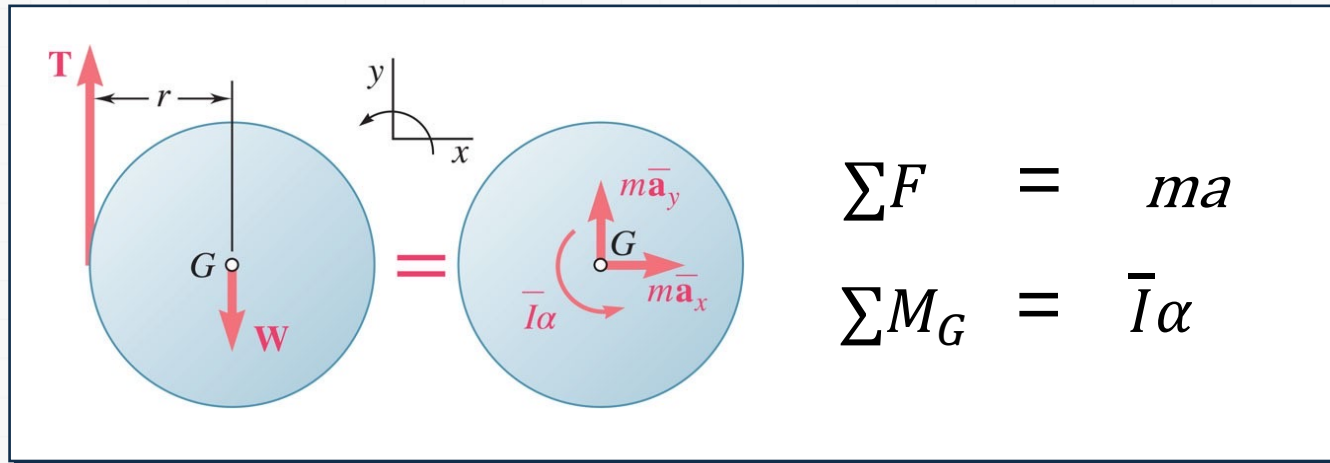


- Isolate the body of interest (free body).
- Draw your axis system (example: Cartesian, polar, path).
- Add in applied forces (example: weight).
- Replace supports with forces (example: tension force).
- Draw appropriate dimensions (angles and distances).

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Free Body Diagrams and Kinetic Diagrams

- ⚙️ Put the inertial terms for the body of interest on the kinetic diagram.
- Isolate **the body of interest** (free body)

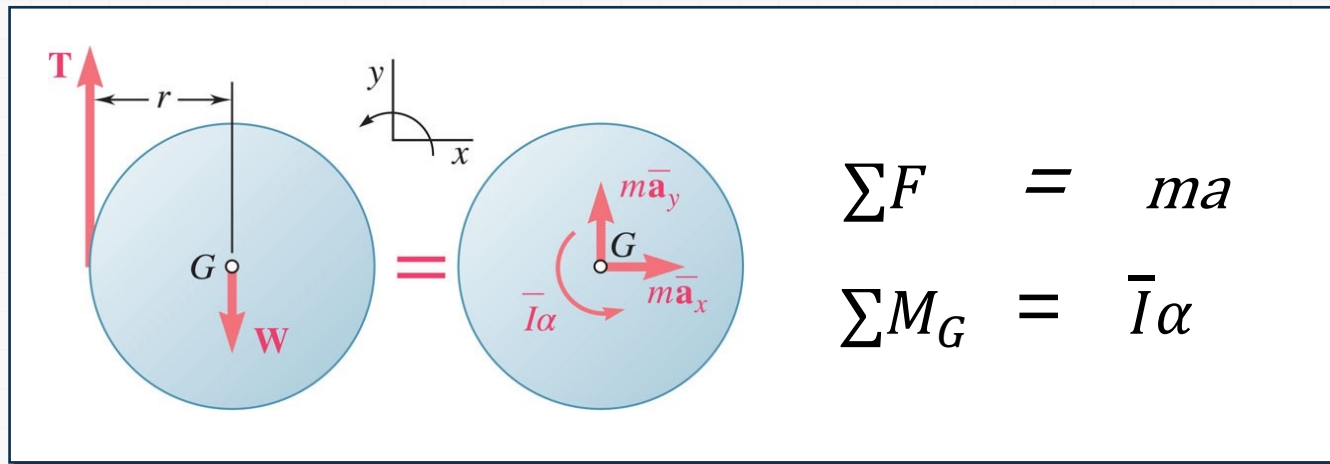


## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Free Body Diagrams and Kinetic Diagrams

⚙️ Put the inertial terms for the body of interest on the kinetic diagram.

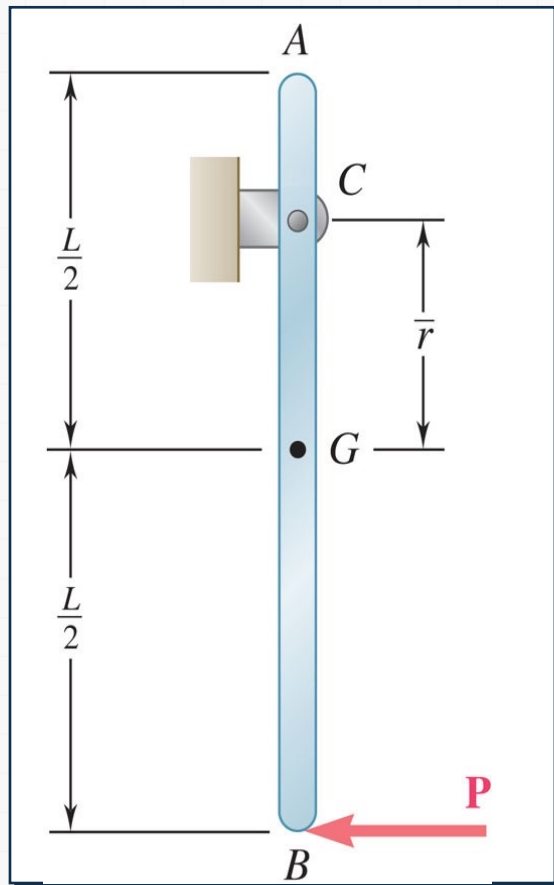
- Draw in the mass times acceleration of the particle; if unknown, do this in the **positive direction according to your chosen axes**. For rigid bodies, also include the rotational term,  $I_G \alpha$



## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Free Body Diagrams and Kinetic Diagrams

⚙️ Draw the FBD and KD for the bar AB of mass  $m$

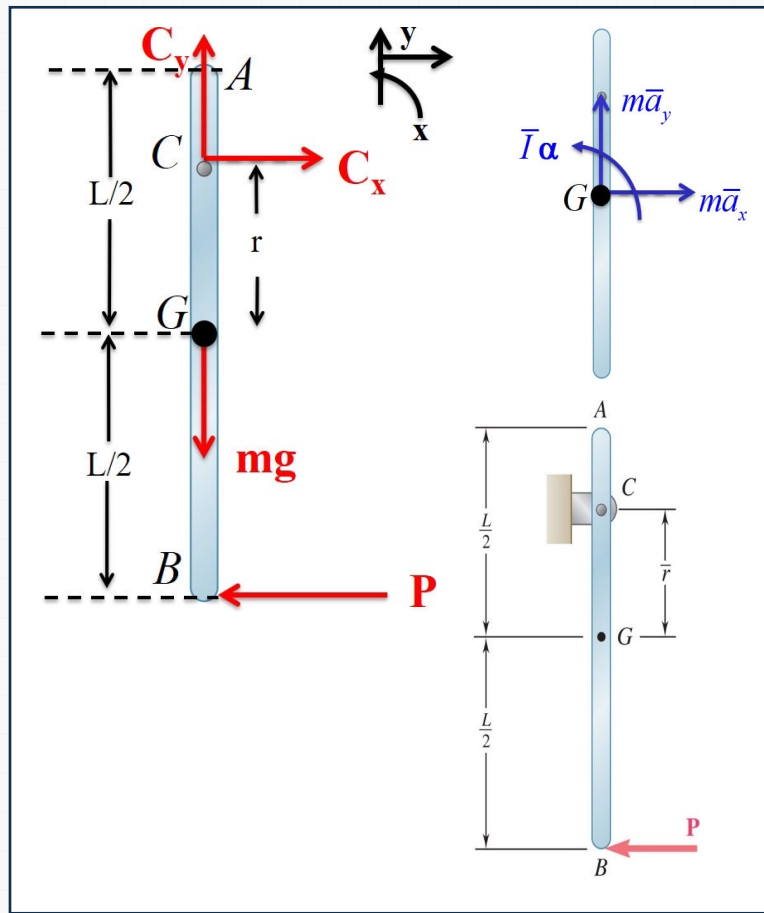


- A known force  $P$  is applied at the bottom of the bar.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Free Body Diagrams and Kinetic Diagrams

⚙️ Draw the FBD and KD for the bar AB of mass  $m$

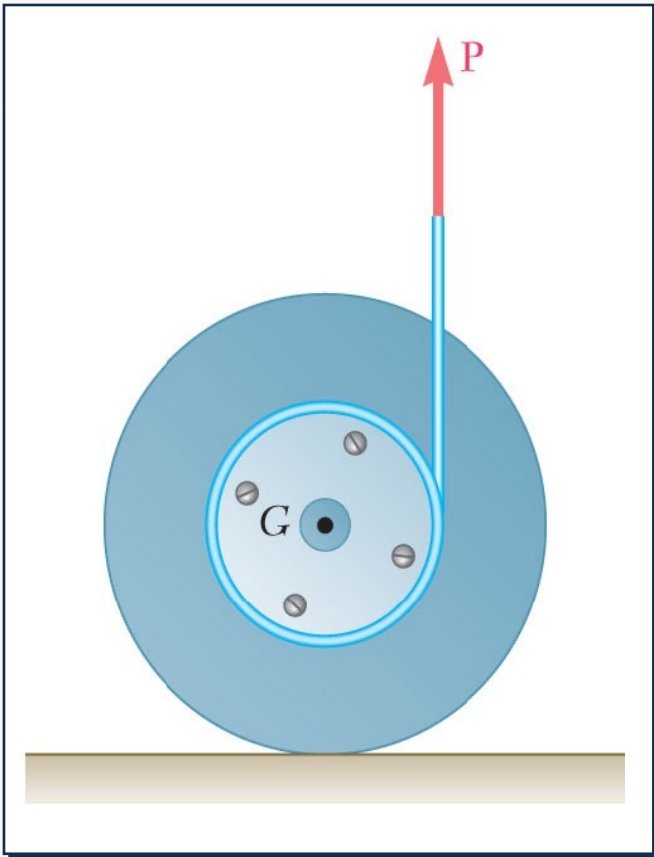


- Isolate body
- Axes
- Applied forces
- Replace supports with forces
- Dimensions
- Kinetic diagram

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Free Body Diagrams and Kinetic Diagrams

⚙️ Draw the FBD and KD for the wheel.

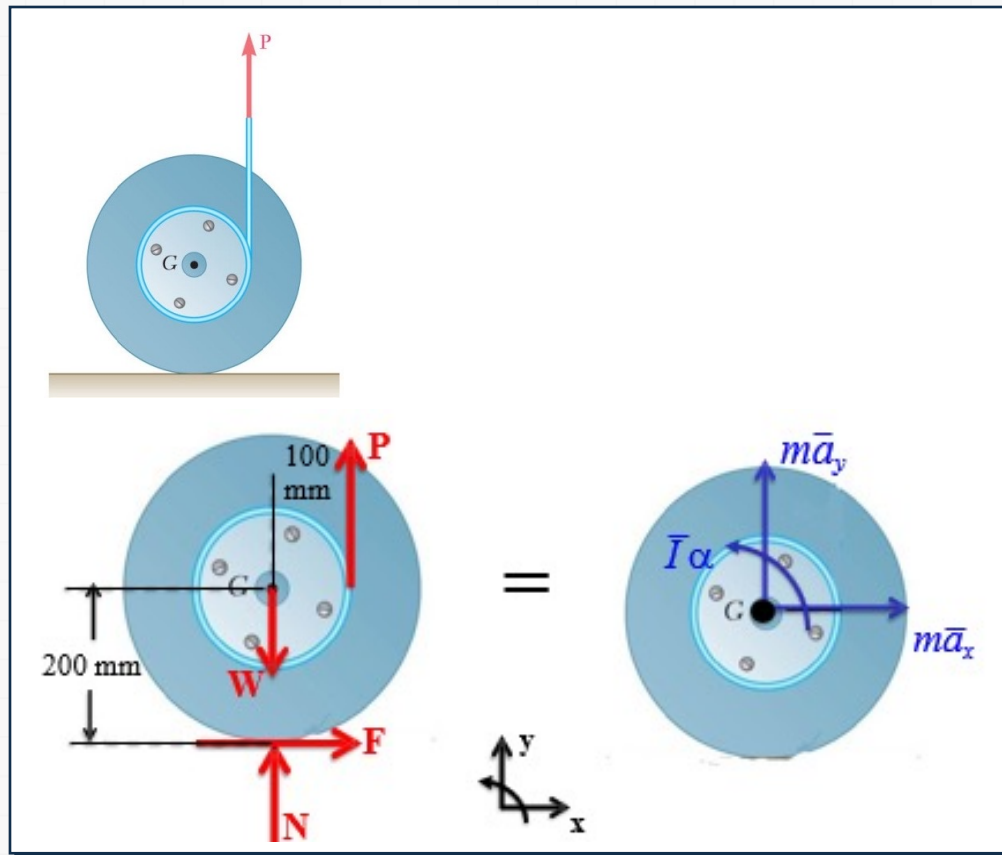


- A drum of 100mm radius is attached to a disk of 200mm radius.
  - The combined drum and disk had a combined mass of 5kg.
  - A cord is attached as shown, and a force of magnitude  $P=25\text{N}$  is applied.
  - The coefficients of static and kinetic friction between the wheel and ground are  $\mu_s = 0.25$  and  $\mu_k = 0.20$ .

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Free Body Diagrams and Kinetic Diagrams

⚙️ Draw the FBD and KD for the wheel.

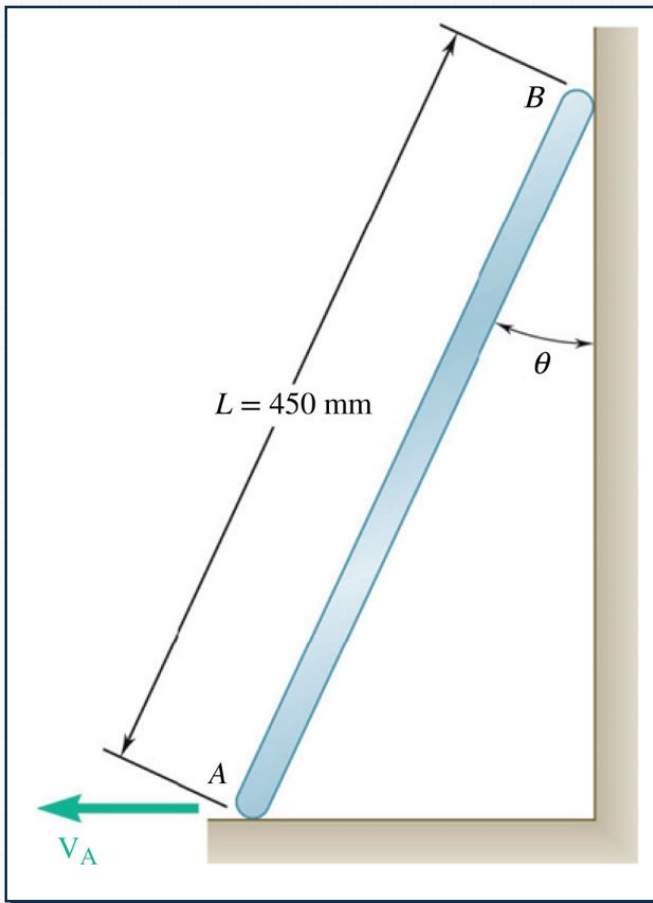


- Isolate body
- Axes
- Applied forces
- Replace supports with forces
- Dimensions
- Kinetic diagram

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Free Body Diagrams and Kinetic Diagrams

⚙️ Draw the FBD and KD for the ladder.

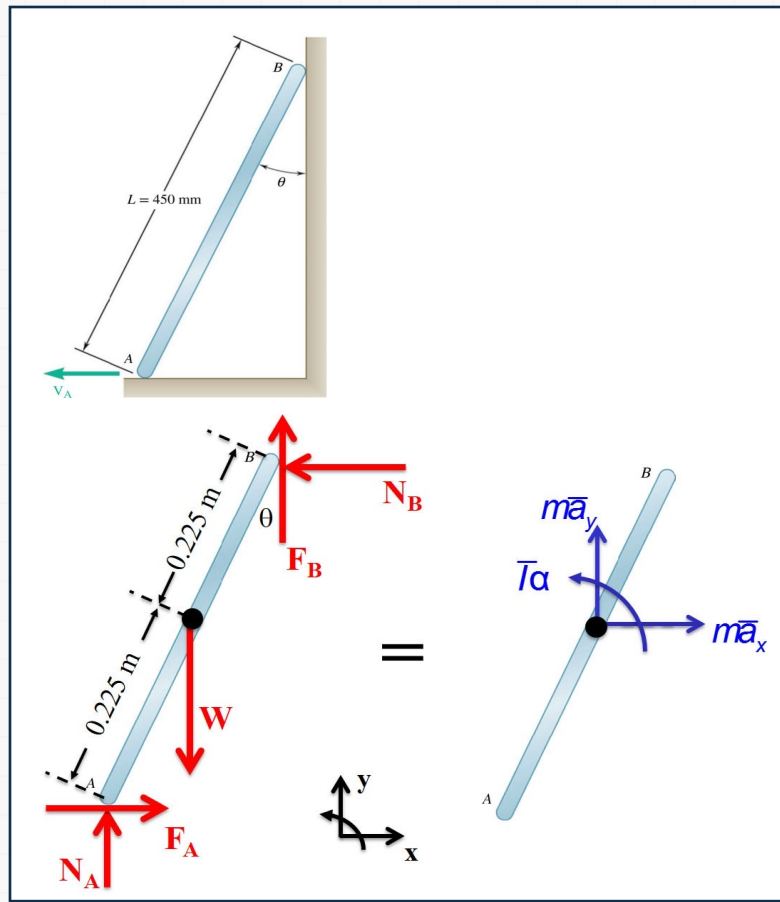


- The ladder AB slides down the wall as shown.
- The wall and floor are both rough.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Free Body Diagrams and Kinetic Diagrams

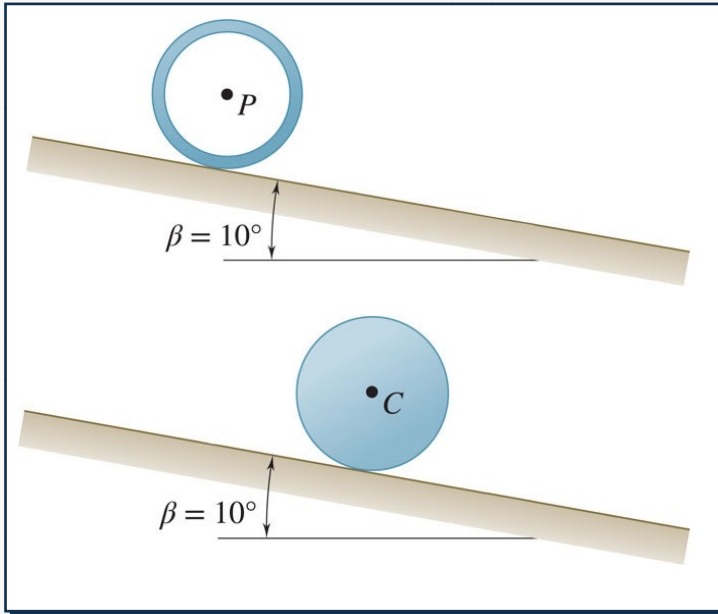
⚙️ Draw the FBD and KD for the ladder.



- Isolate body
- Axes
- Applied forces
- Replace supports with forces
- Dimensions
- Kinetic diagram

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Concept Question

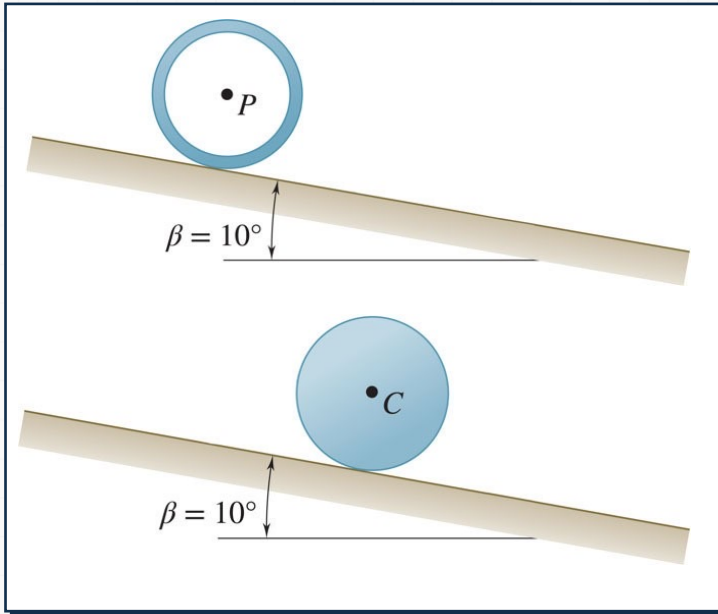


• The thin pipe **P** and the uniform cylinder **C** have the same outside radius and the same mass. If they are both released from rest, **which of the following statements is true?**

- a) The pipe P will have a greater acceleration
- b) The cylinder C will have a greater acceleration
- c) The cylinder and pipe will have the same acceleration

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Concept Question



• The thin pipe **P** and the uniform cylinder **C** have the same outside radius and the same mass. If they are both released from rest, **which of the following statements is true?**

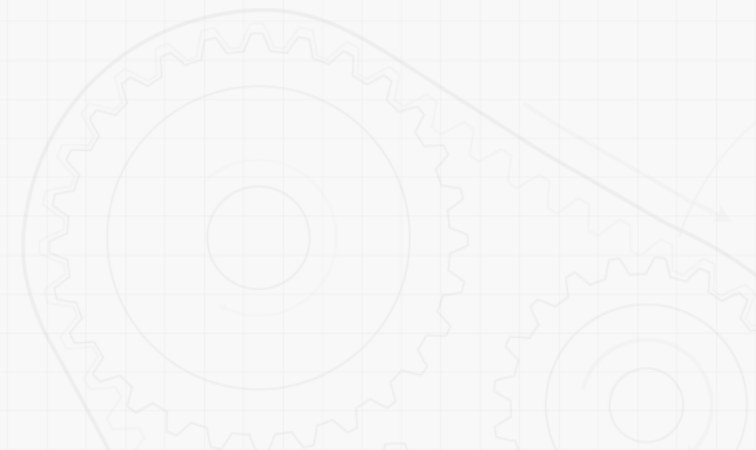
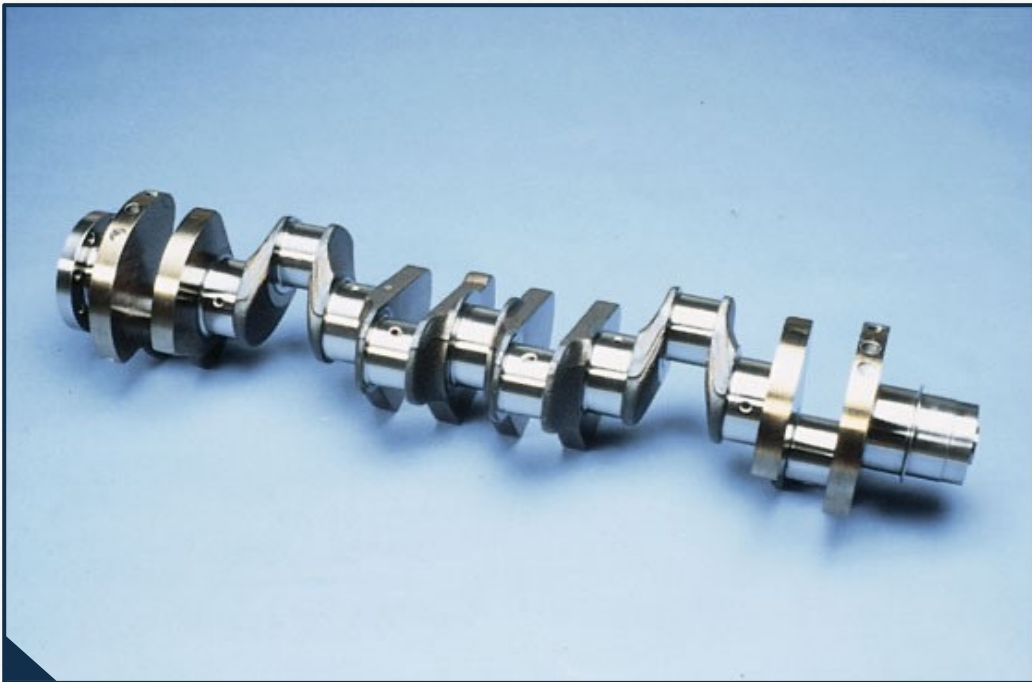
- a) The pipe P will have a greater acceleration
- **b) The cylinder C will have a greater acceleration**
- c) The cylinder and pipe will have the same acceleration



## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

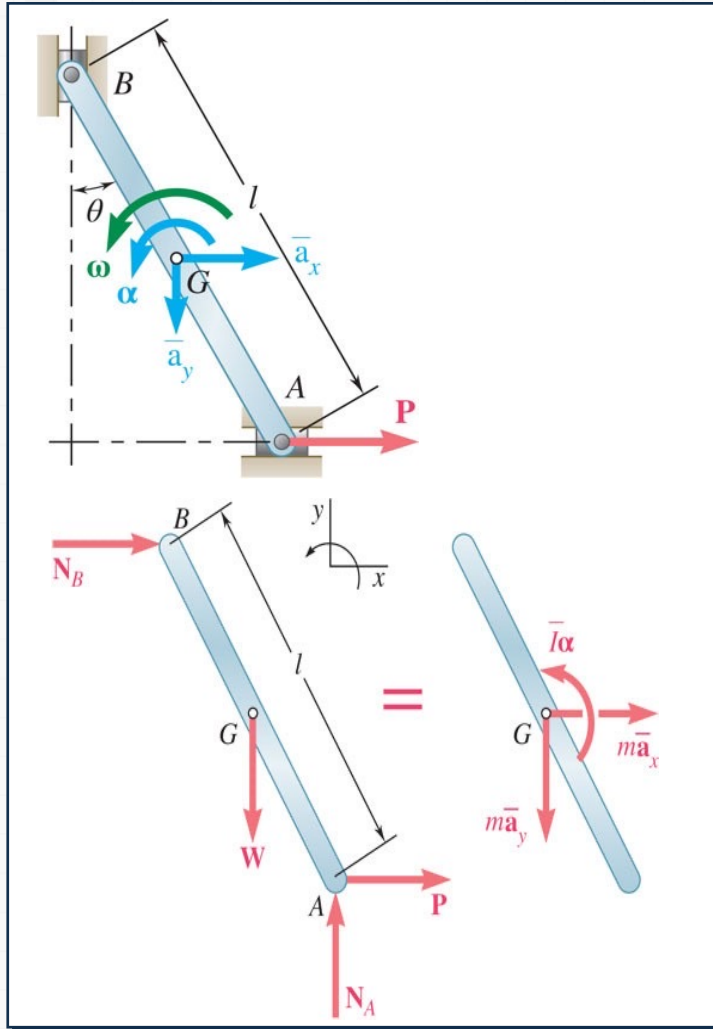
# Constrained Plane Motion

The reactions of the automobile crankshaft bearings depend on the mass, mass moment of inertia, and the kinematics of the crankshaft.



## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

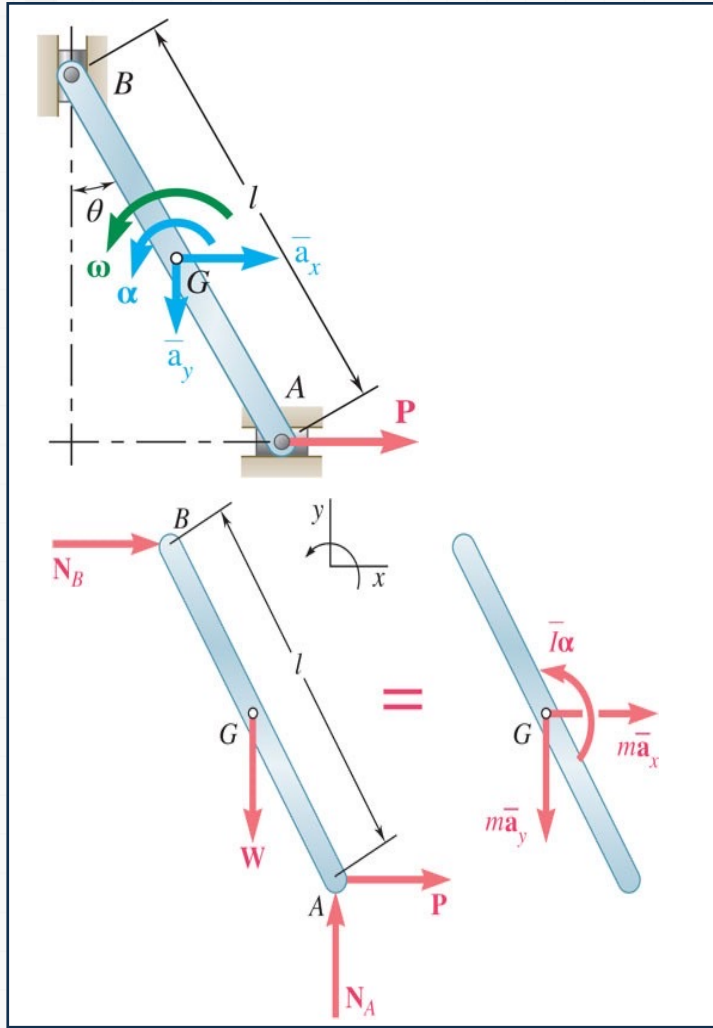
# Constrained Plane Motion



- Most engineering applications involve rigid bodies which are moving under given constraints,
  - example: cranks, connecting rods, and non-slipping wheels.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Constrained Plane Motion

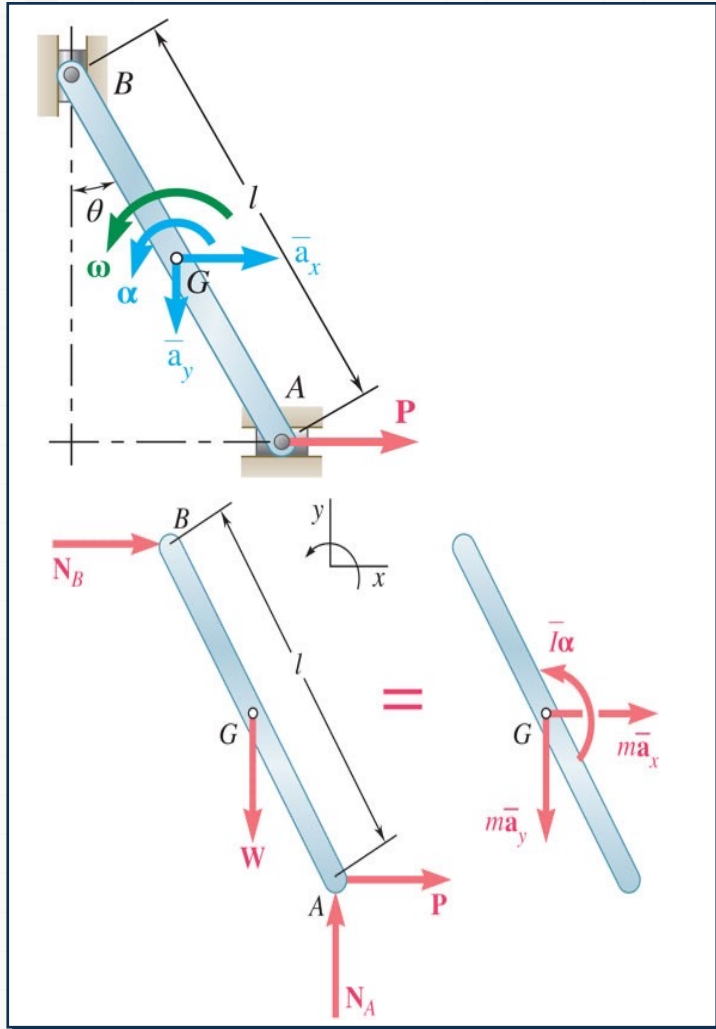


- Constrained plane motion

- motions with definite relations between the components of acceleration of the mass center and the angular acceleration of the body.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

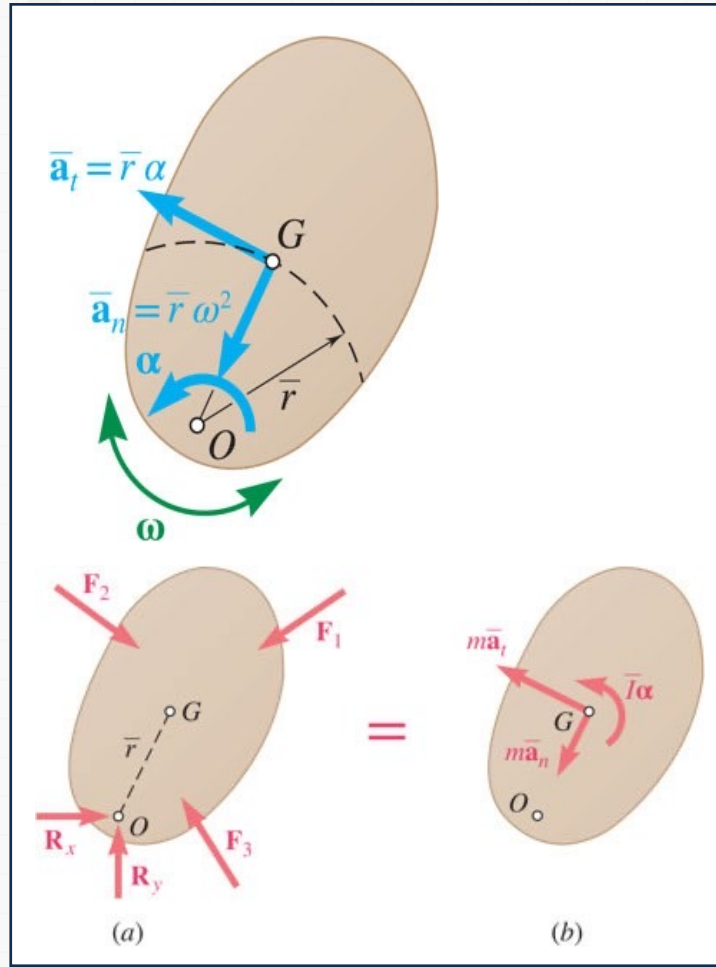
# Constrained Plane Motion



- Solution of a problem involving constrained plane motion begins with a kinematic analysis.
- Example: given  $\theta$ ,  $\omega$ , and  $\alpha$ , find  $P$ ,  $N_A$ , and  $N_B$ .
  - kinematic analysis yield  $\bar{a}_x$  and  $\bar{a}_y$ .
  - We can then find  $P$ ,  $N_A$  and  $N_B$  by solving the appropriate equations.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Constrained Motion: Noncentroidal Rotation



- Noncentroidal rotation

- motion of a body is constrained to rotate about a fixed axis that does not pass through its mass center.

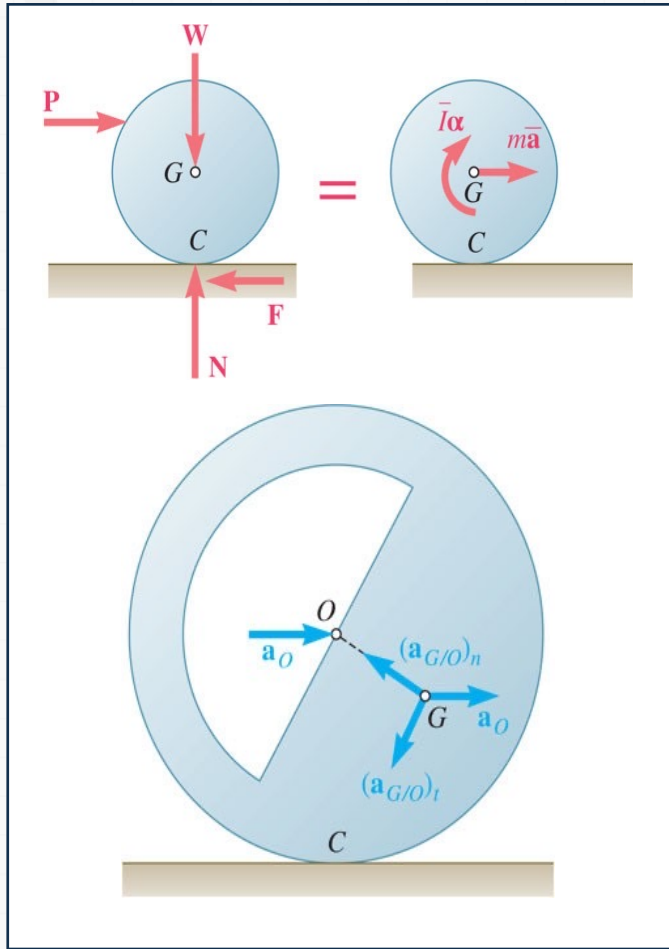
- Kinematic relation between the motion of the mass center  $G$  and the motion of the body about  $G$ ,

$$\bar{a}_t = \bar{r}\alpha$$

$$\bar{a}_n = \bar{r}\omega^2$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Constrained Plane Motion: Rolling Motion



- For a balanced disk constrained to roll without sliding,

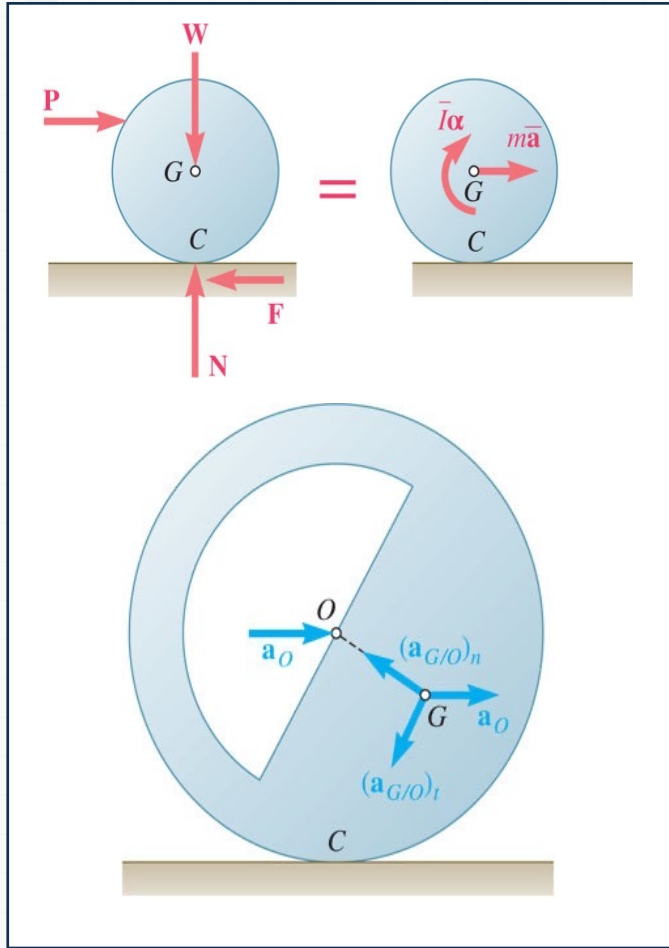
$$\bar{x} = r\theta \rightarrow \bar{a} = r\alpha$$

- Rolling, no sliding

$$F \leq \mu_s N \quad \bar{a} = r\alpha$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Constrained Plane Motion: Rolling Motion



- Rolling, sliding impending

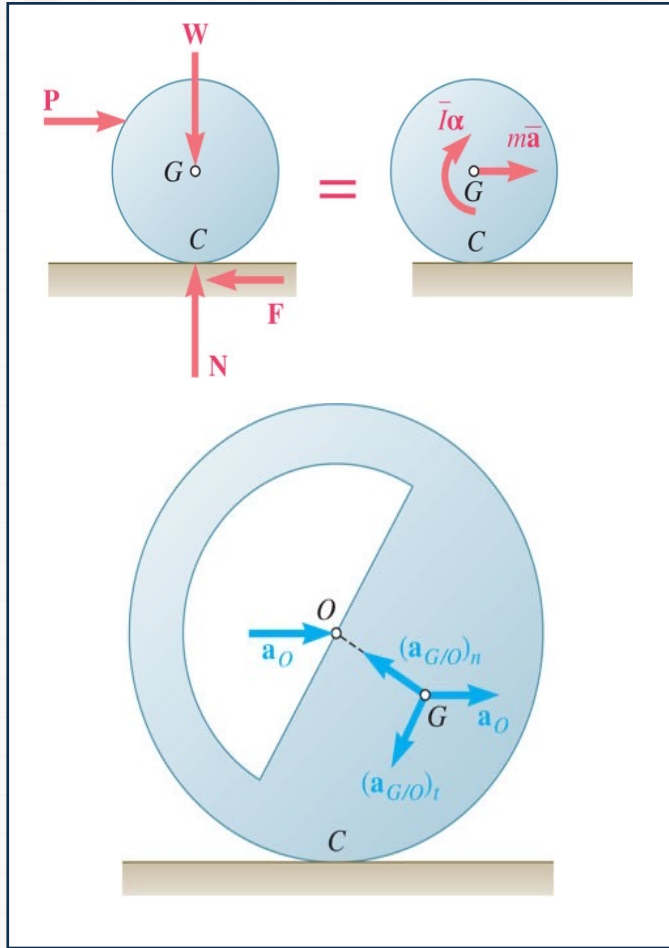
$$F = \mu_s N \quad \bar{a} = r\alpha$$

- Rotating and sliding

$$F = \mu_k N \quad \bar{a}, r\alpha \text{ independent}$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Constrained Plane Motion: Rolling Motion



- For the geometric center of an unbalanced disk

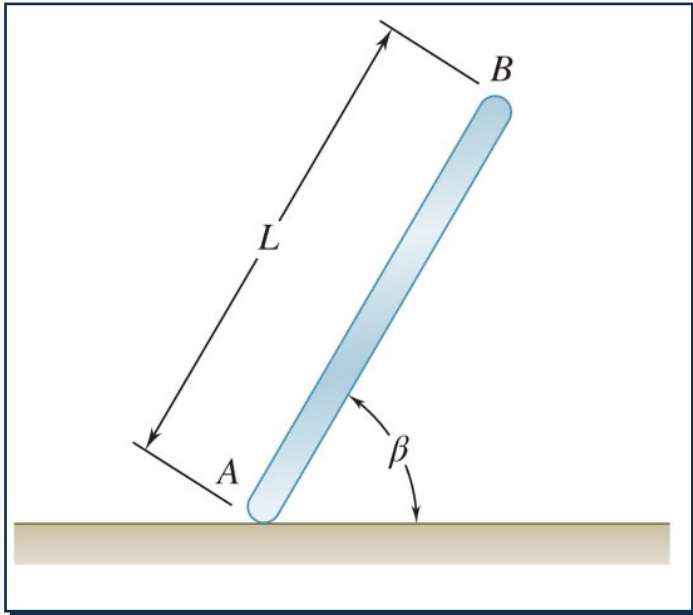
$$a_O = r\alpha$$

- The acceleration of the mass center,

$$\begin{aligned}\vec{a}_G &= \vec{a}_O + \vec{a}_{G/O} \\ &= \vec{a}_O + (\vec{a}_{G/O})_t + (\vec{a}_{G/O})_n\end{aligned}$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

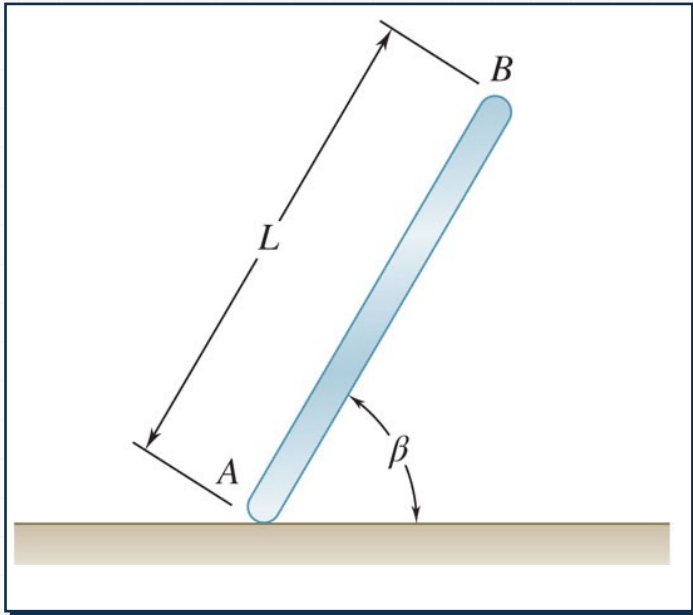
# Concept Question



- What would be true if the floor was smooth and friction was zero?
  - a) The bar would rotate about point A
  - b) The bar's center of gravity would go straight downwards
  - c) The bar would not have any angular acceleration

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

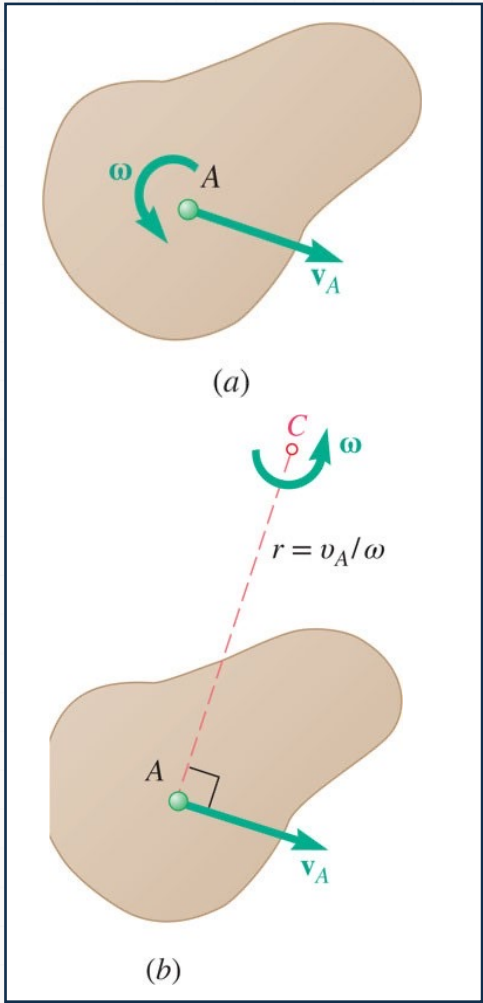
# Concept Question



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## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

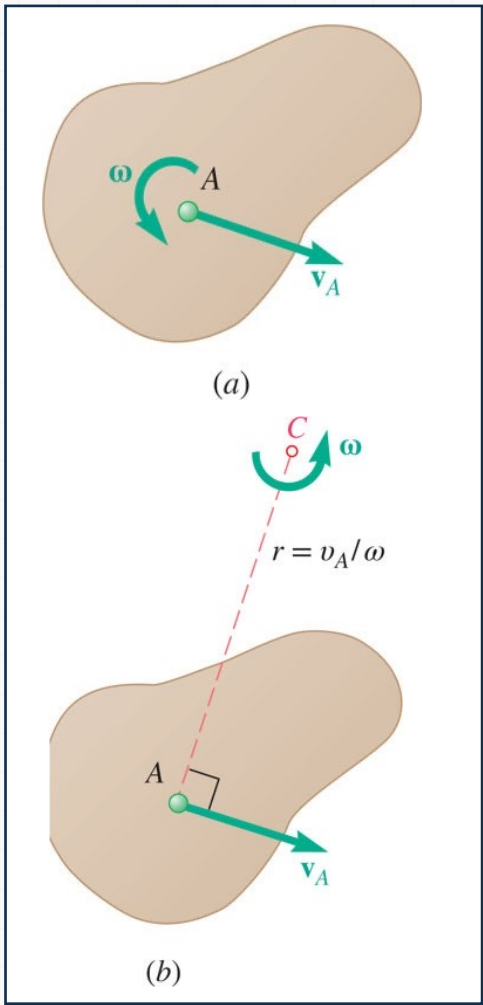
# Instantaneous Center of Rotation



- Plane motion of all particles in a slab
  - The transformation of any point A and the rotation to A at an angular velocity independent of the selection of A can always be replaced.
- The same translational and rotational velocities at A
  - be obtained by allowing the slab to rotate with the same angular velocity about the point C on a perpendicular to the velocity at A.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

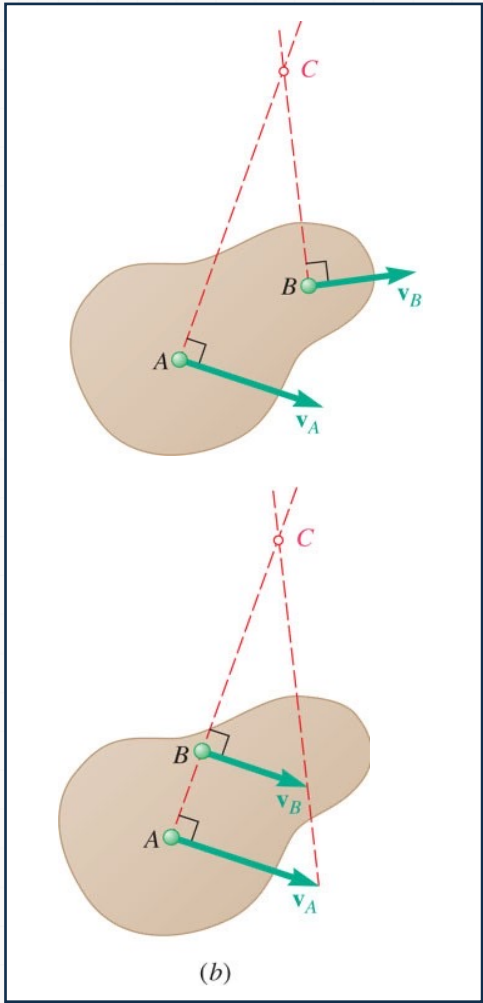
# Instantaneous Center of Rotation



- Velocity of all other particles in the slab
  - The same as originally defined since the angular velocity and translational velocity at A are equivalent.
- As far as the velocities
  - The slab seems to rotate about the instantaneous center of rotation C.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

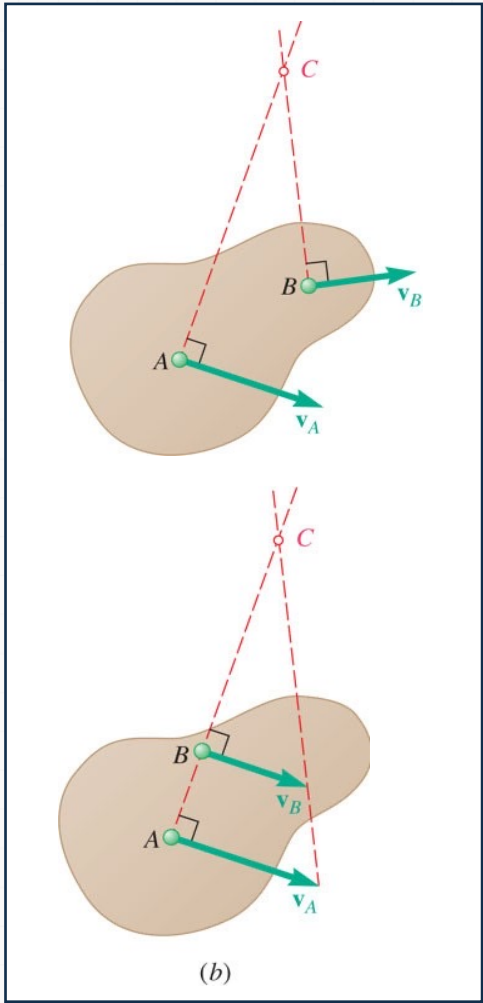
# Instantaneous Center of Rotation



- If the velocity at **two points A and B** are known,
  - The instantaneous center of rotation lies at the intersection of the perpendiculars to the velocity through A and B.
- If the velocity vectors are **parallel**,
  - The instantaneous center of rotation is at infinity and the angular velocity is zero.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

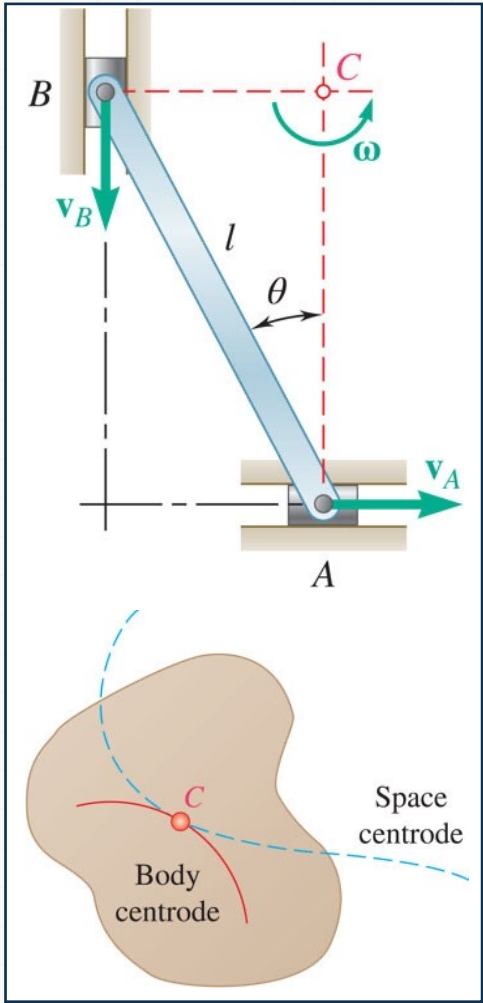
# Instantaneous Center of Rotation



- If the velocity vectors at A and B are perpendicular to the line AB,
  - The instantaneous center of rotation lies at the intersection of the line AB with the line joining the extremities of the velocity vectors at A and B.
- If the velocity magnitudes are equal,
  - The instantaneous center of rotation is at infinity and the angular velocity is zero.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Instantaneous Center of Rotation



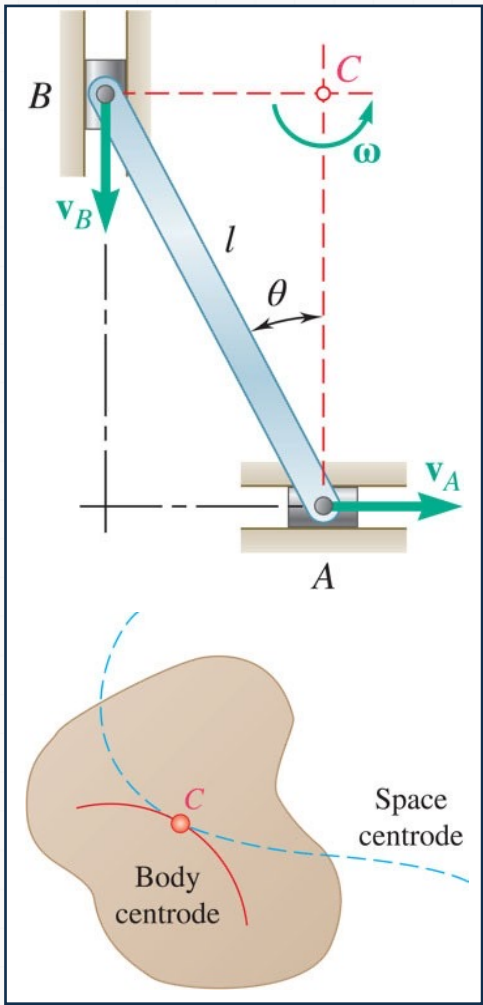
- The instantaneous center of rotation lies at the intersection of the perpendiculars to the velocity vectors through A and B.

$$\omega = \frac{v_A}{AC} = \frac{v_A}{l \cos \theta}$$

$$v_B = (BC)\omega = (l \sin \theta) \frac{v_A}{l \cos \theta} = v_A \tan \theta$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

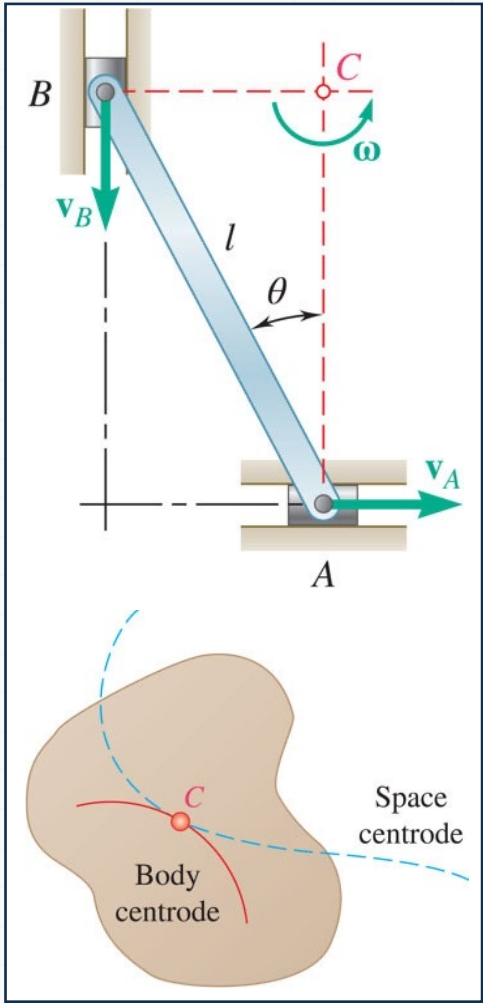
# Instantaneous Center of Rotation



- The velocities of all particles on the rod
  - be as if they were rotated about  $C$
- The particle at the center of rotation
  - Velocity has zero
  - The particle coinciding with the center of rotation changes with time.
  - The acceleration of the particle at the instantaneous center of rotation is not zero.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

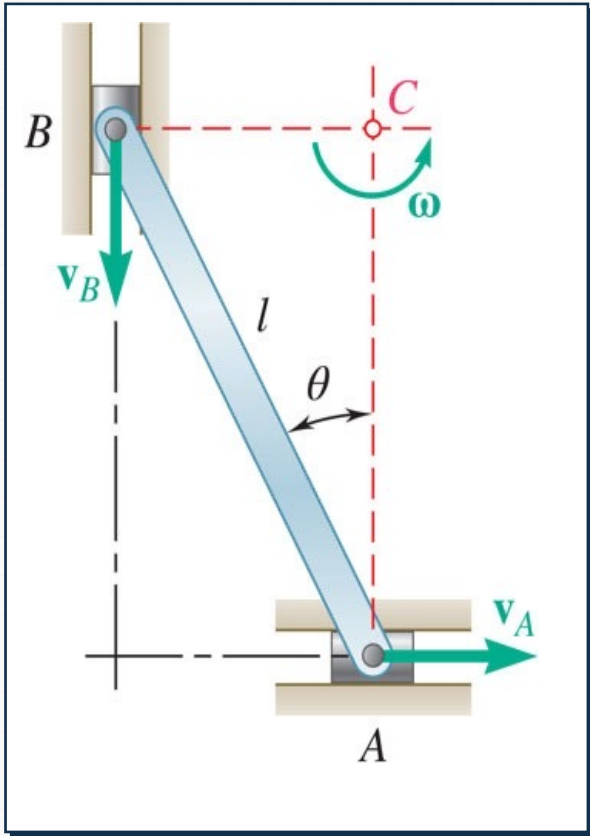
# Instantaneous Center of Rotation



- The acceleration of the particles in the slab
  - It cannot be determined as if the slab were simply rotating about  $C$ .
- The trace of the locus of the center of rotation on the body is the body centre and in space is the space centre.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Instantaneous Center of Rotation

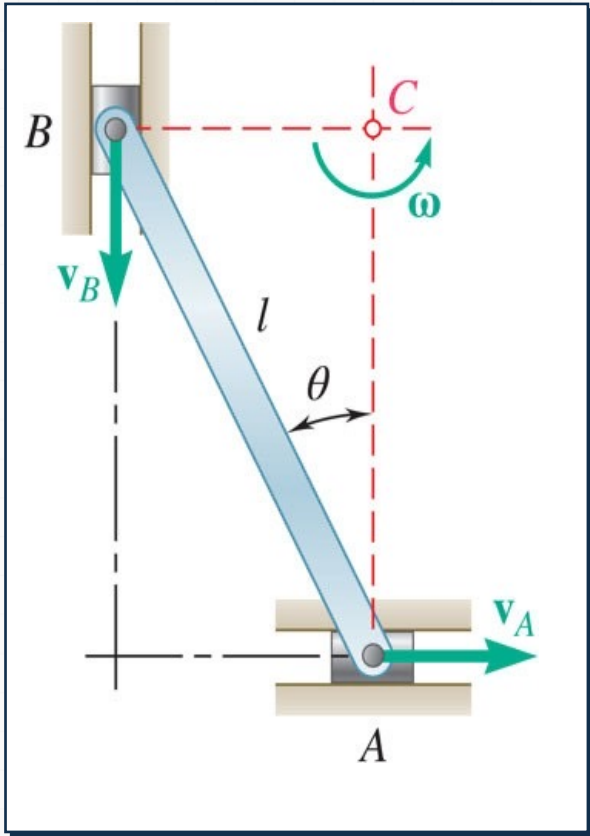


• At the instant shown, what is the approximate direction of the velocity of point G, the center of bar AB?

- a)  $\rightarrow$
- b)  $\nwarrow$
- c)  $\swarrow$
- d)  $\downarrow$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Instantaneous Center of Rotation



• At the instant shown, what is the approximate direction of the velocity of point G, the center of bar AB?

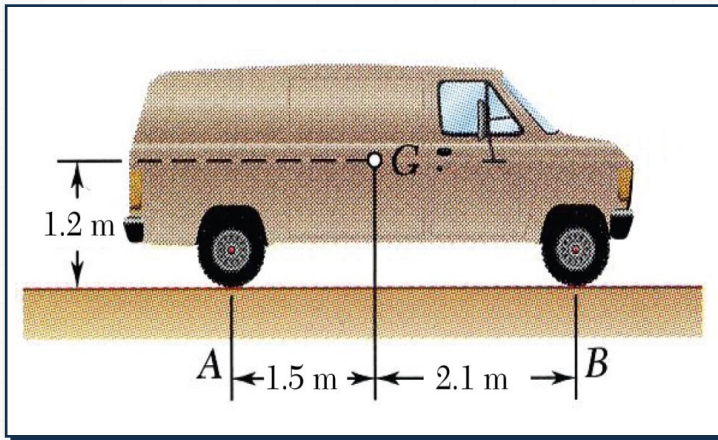
- a)  $\rightarrow$
- b)  $\nwarrow$
- c)  $\searrow$
- d)  $\downarrow$



## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 1

⚙️ **The truck brakes worked and the wheels stopped spinning**

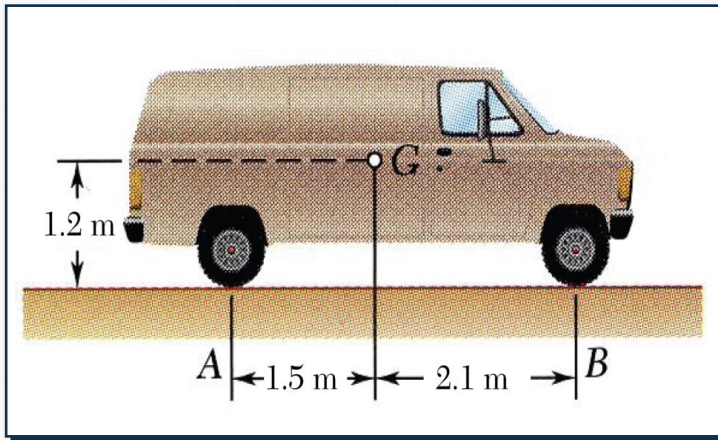


- At a forward speed of  $10\text{ m/s}$ , the truck brakes were applied, causing the wheels to stop rotating.
- It was observed that the truck to skidded to a stop in  $7\text{ m}$ .

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 1

⚙️ **The truck brakes worked and the wheels stopped spinning**



- Determine
  - The magnitude of the normal reaction
  - The friction force at each wheel as the truck skidded to a stop.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 1

## ⚙ The truck brakes worked and the wheels stopped spinning

### • Strategy

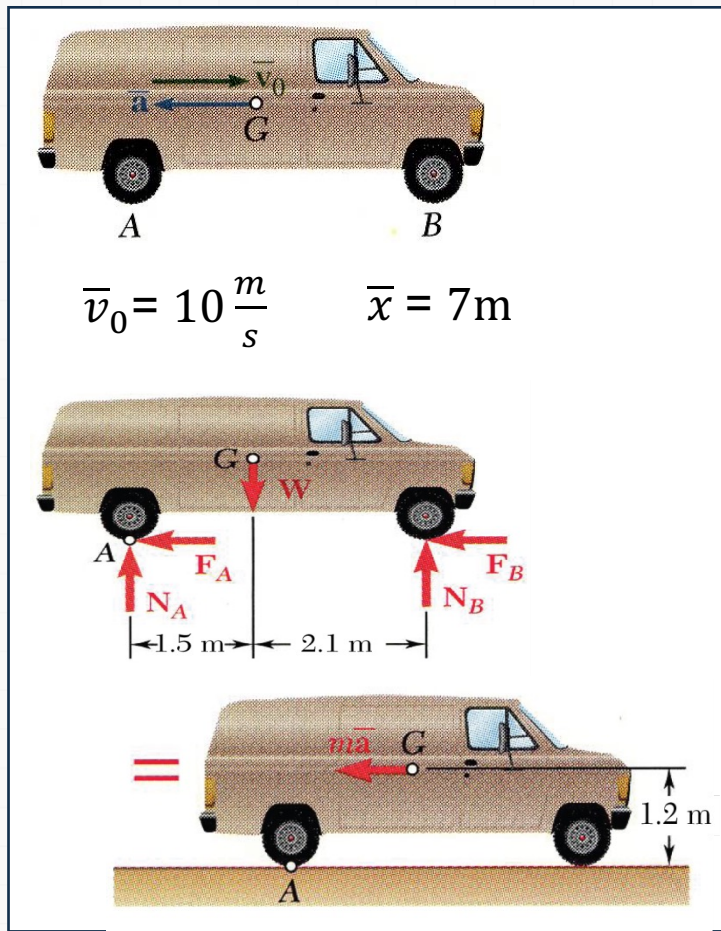
- Calculate the acceleration during the skidding stop by assuming uniform acceleration.
- Draw the free-body-diagram equation expressing the equivalence of the external and effective forces.
- Apply the three corresponding scalar equations to solve for the unknown normal wheel forces at the front and rear and the coefficient of friction between the wheels and road surface.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 1

## Modeling and Analysis

- Calculate the acceleration during the skidding stop by assuming uniform acceleration.



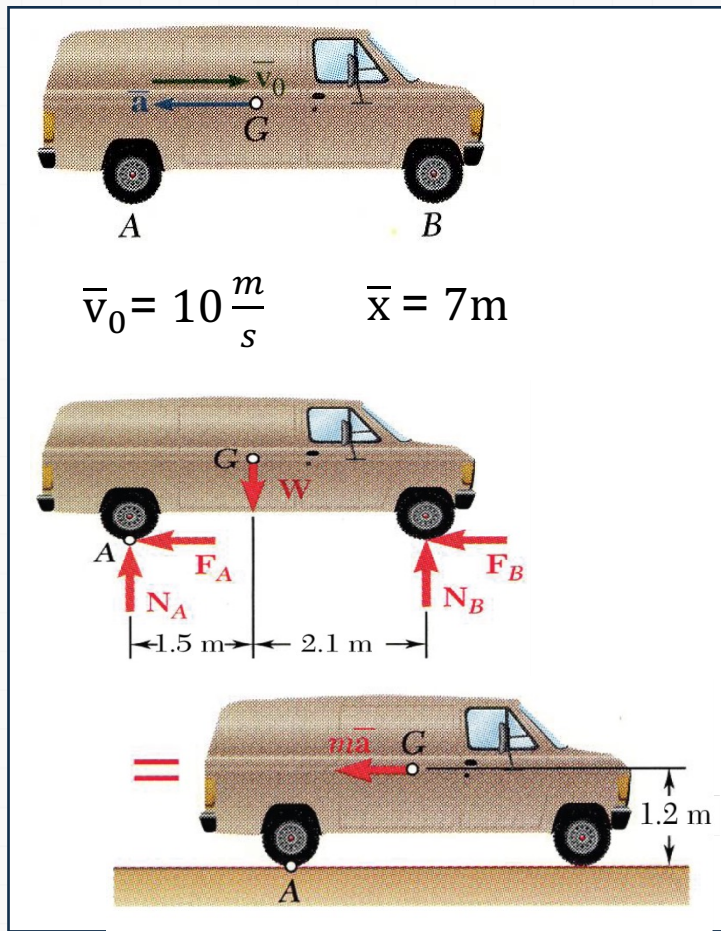
$$\bar{v}^2 - \bar{v}_0^2 = -2\bar{a}(\bar{x} - \bar{x}_0)$$

$$0 = (10)^2 - 2\bar{a}(7) \quad \bar{a} = 7.14 \frac{\text{m}}{\text{s}^2}$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 1

## ⚙ Modeling and Analysis



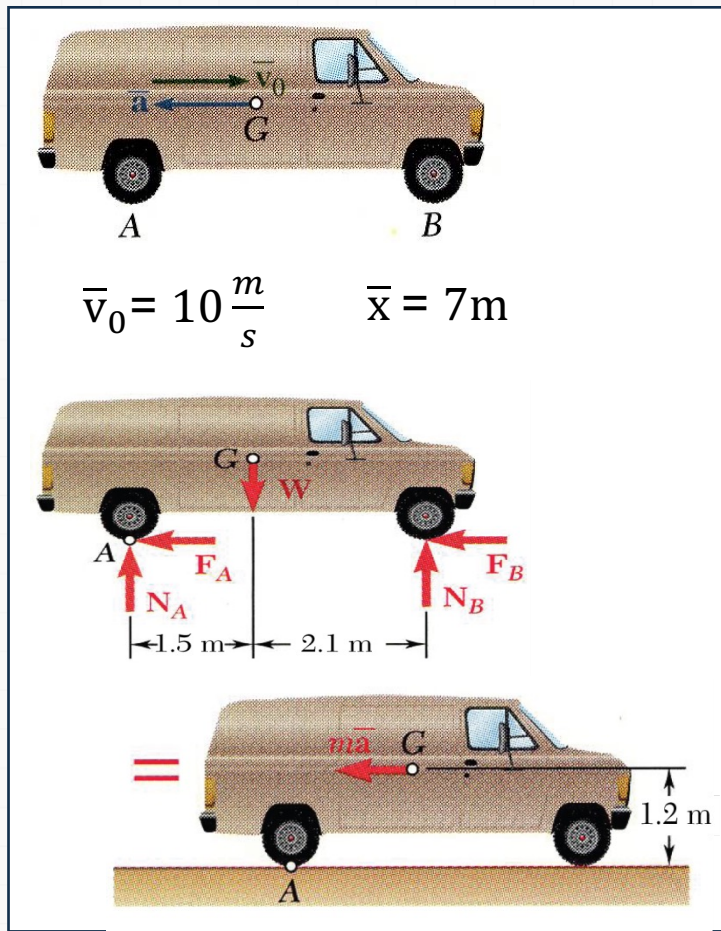
- Draw a **free-body-diagram equation** expressing the equivalence of the external and inertial terms.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 1

## Modeling and Analysis

- Apply the corresponding scalar equations.



$$+\uparrow \sum F_y = \sum (F_y)_{eff} \quad N_A + N_B - W = 0$$

$$+\rightarrow \sum F_x = \sum (F_x)_{eff} \quad -F_A - F_B = -m\bar{a}$$

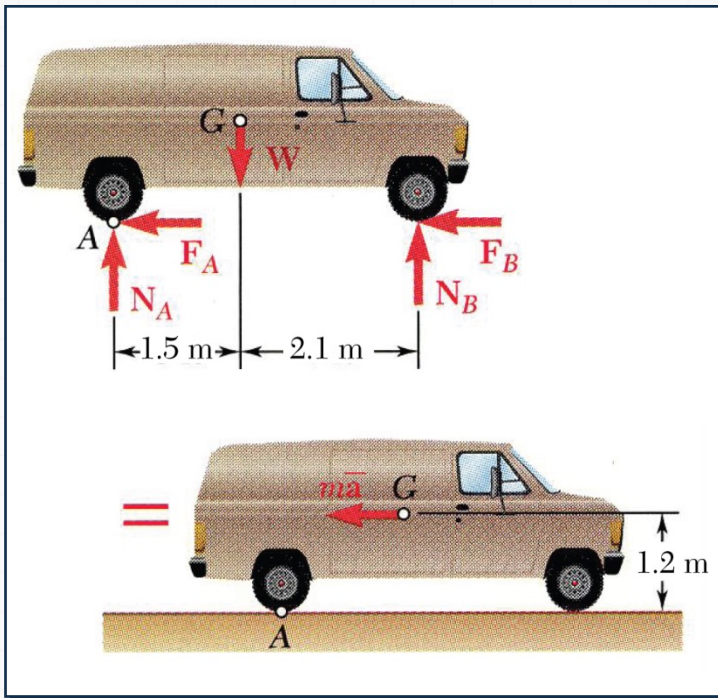
$$-\mu_k(N_A + N_B) = -\mu_k W = -(m)\bar{a}$$

$$m_k = \frac{\bar{a}}{g} = \frac{7.14}{9.81} = 0.728$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 1

## ⚙ Modeling and Analysis



- Apply the corresponding scalar equations.

$$+\circlearrowleft \sum M_A = \sum (M_A)_{eff}$$

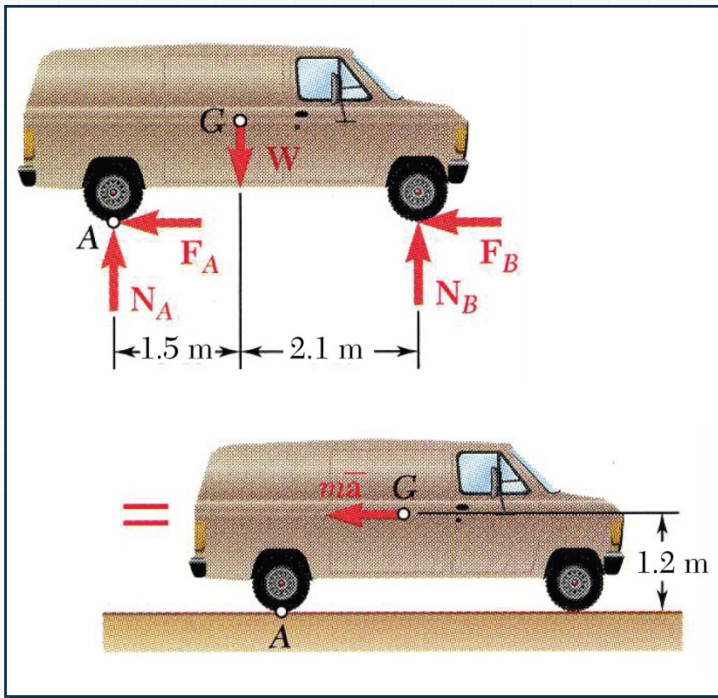
$$-(1.5)W + (3.6)N_B = (1.2)m\bar{a}$$

$$N_B = \frac{1}{3.6} \left( 1.5W + 1.2 \frac{W}{g} \bar{a} \right) = \frac{W}{3.6} \left( 1.5 + 1.2 \frac{\bar{a}}{g} \right)$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 1

## ⚙ Modeling and Analysis



- Apply the corresponding scalar equations.

$$N_B = 0.659W \quad N_A = W - N_B = 0.341W$$

$$N_{rear} = \frac{1}{2}N_A = \frac{1}{2}(0.341W)$$

$$N_{rear} = 0.1705W$$

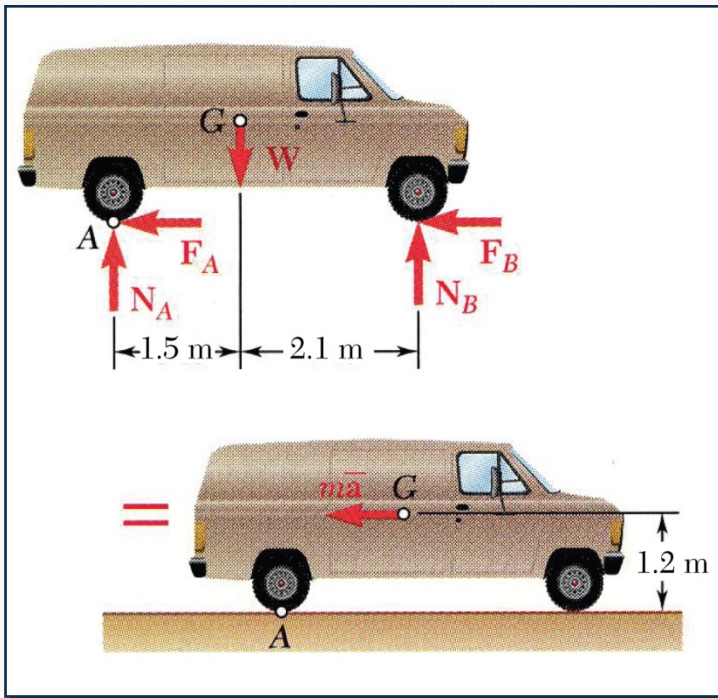
$$F_{rear} = \mu_k N_{rear} = (0.728)(0.1705W)$$

$$F_{rear} = 0.1241W$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 1

## ⚙ Modeling and Analysis



- Apply the corresponding scalar equations.

$$N_{front} = \frac{1}{2} N_V = \frac{1}{2} (0.659W)$$

$$N_{front} = 0.3295W$$

$$F_{front} = \mu_k N_{front} = (0.728)(0.3295W)$$

$$F_{front} = 0.240W$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 1

## ⚙️ Reflect and Think

Note that even though the angular acceleration of the van is zero, the sum of the moments about point A is not equal to zero

since from the kinetic diagram,  $m\vec{a}$  produces a moment about A.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 1

## ⚙️ Reflect and Think

Rather than taking moments about point A, you also could have chosen to take moments about the center of mass, G.

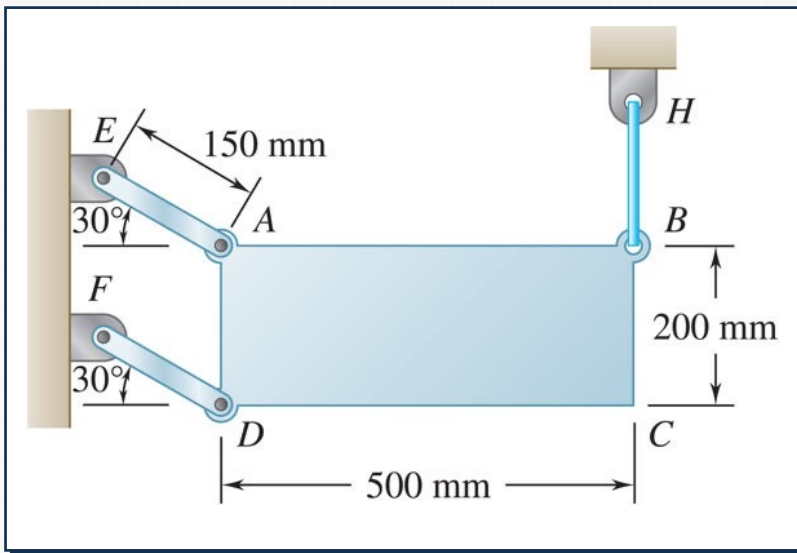
In this case, the sum of the moments would have been equal to zero.

You only get three independent equations for a rigid body in plane motion:  $\Sigma F_x$ ,  $\Sigma F_y$ , and one moment equation.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

## Sample Problem 2

⚙ The thin plate of mass 8kg is held in place as shown.



- Neglecting the mass of the links, determine immediately after the wire has been cut
  - a) the acceleration of the plate
  - b) the force in each link.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

## Sample Problem 2

⚙️ **The thin plate of mass 8kg is held in place as shown.**

• **Strategy**

- Note that after the wire is cut, all particles of the plate move along parallel circular paths of radius 150mm. The plate is in curvilinear translation.
- Draw **the free-body-diagram equation** expressing the equivalence of the external and effective forces.

**13 Plane Motion of Rigid Bodies: Forces and Accelerations**

# Sample Problem 2

⚙️ **The thin plate of mass 8kg is held in place as shown.**

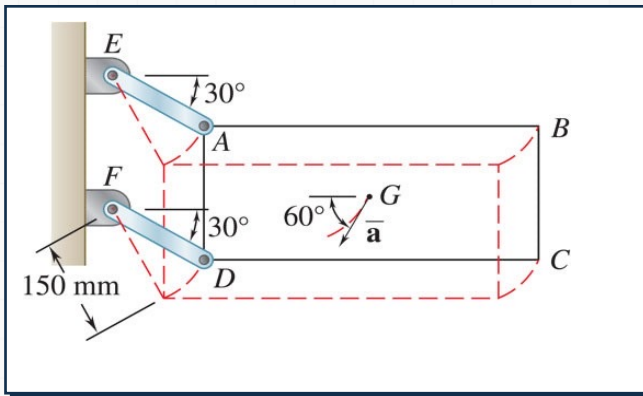
• **Strategy**

- Resolve into **scalar component equations** parallel and perpendicular to the path of the mass center.
- Solve **the component equations and the moment equation** for the unknown acceleration and link forces.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 2

## ⚙ Modeling and Analysis

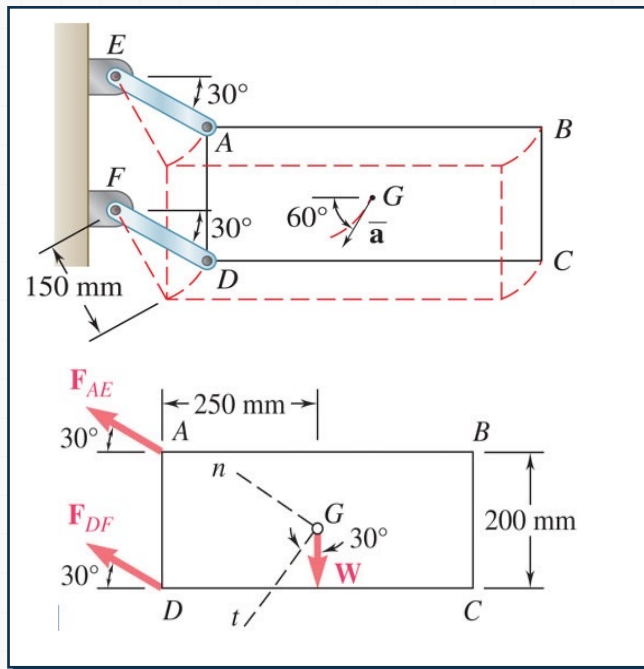


- Note that after the wire is cut, all particles of the plate move along **parallel circular paths of radius 150mm**.
  - The plate is in **curvilinear translation**.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 2

## ⚙ Modeling and Analysis

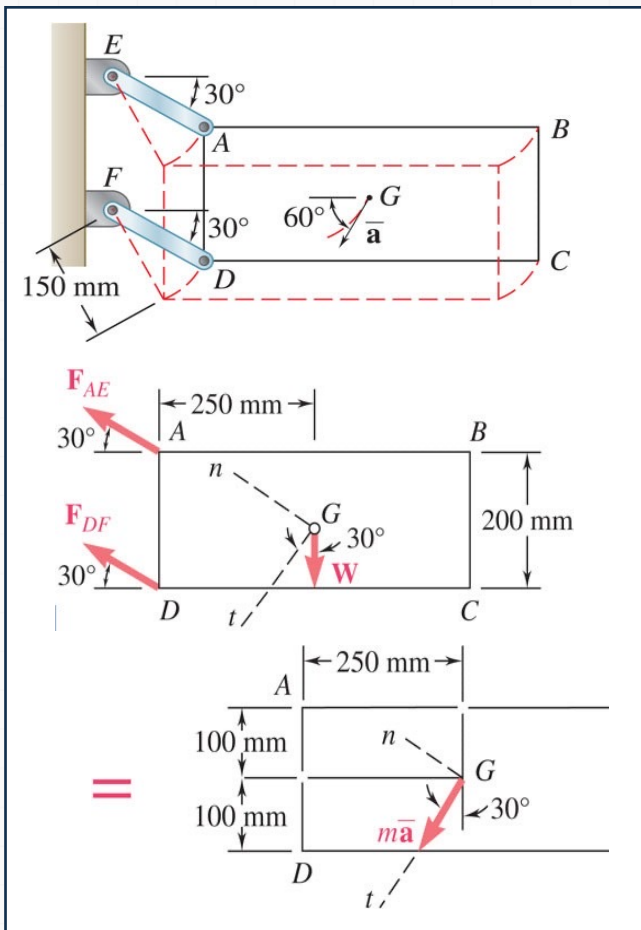


- Draw the free-body-diagram equation expressing the equivalence of the external and effective forces.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

## Sample Problem 2

## ⚙ Modeling and Analysis



- Resolve the diagram equation into components parallel and perpendicular to the path of the mass center.

$$+\swarrow \sum F_t = m\bar{a}_t$$

$$W \cos 30^\circ = m\bar{a}$$

$$mg \cos 30^\circ = m\bar{a}$$

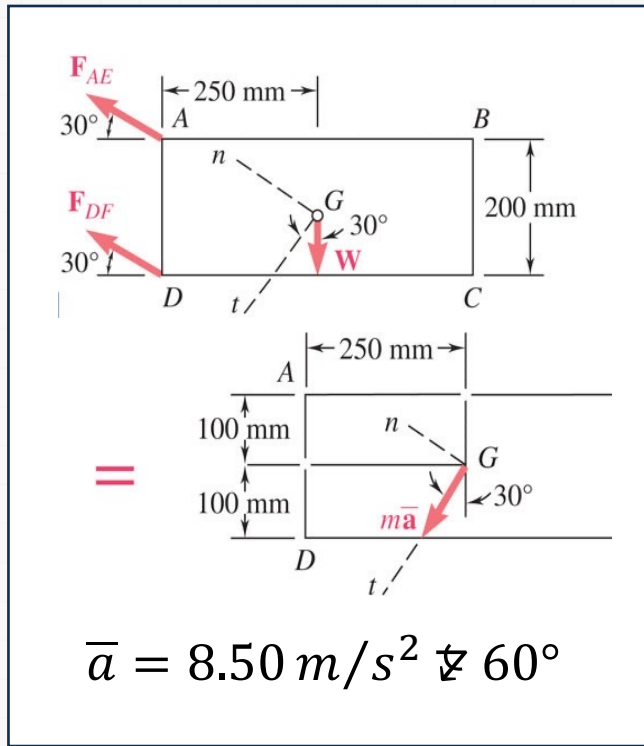
$$\bar{a} = (9.81) \cos 30^\circ$$

$$\bar{a} = 8.50 \text{ m/s}^2 \nlessgtr 60^\circ$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 2

## Modeling and Analysis



- Solve the component equations and the moment equation for the unknown acceleration and link forces.

$$+\circlearrowleft \sum M_G = \bar{I} \alpha$$

$$-(F_{AE} \sin 30^\circ)(250) + (F_{AE} \cos 30^\circ)(100) + \\ -(F_{DF} \sin 30^\circ)(250) - (F_{DF} \cos 30^\circ)(100) = 0$$

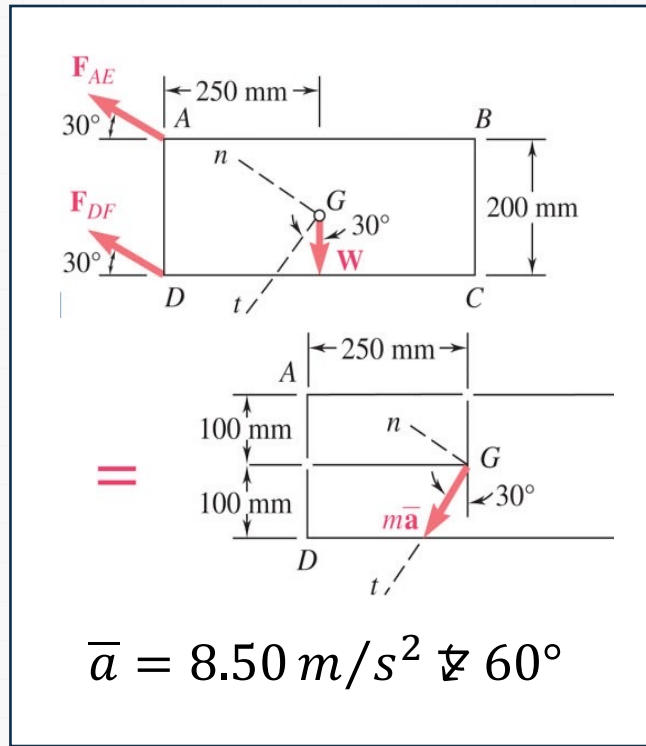
$$38.4 F_{DF} + 211.6 F_{DF} = 0$$

$$F_{DF} = -0.1815 F_{AE}$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 2

## Modeling and Analysis



- Solve the component equations and the moment equation for the unknown acceleration and link forces.

$$+\curvearrowright \sum F_n = m\bar{a}_n$$

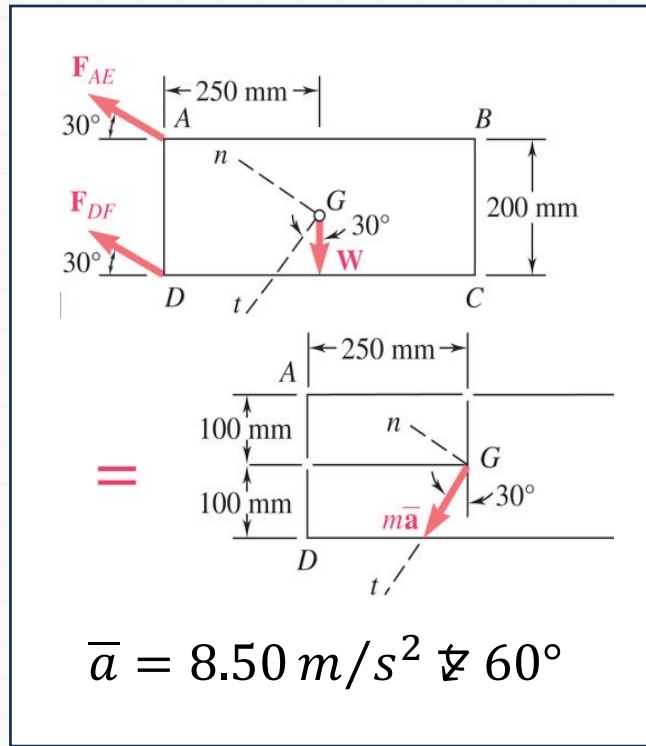
$$F_{AE} + F_{DF} - W \sin 30^\circ = 0$$

$$F_{AE} - 0.1815F_{AE} - W \sin 30^\circ = 0$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 2

## ⚙ Modeling and Analysis



- Solve the component equations and the moment equation for the unknown acceleration and link forces.

$$F_{AE} = 0.619(8)(9.81)$$

$$F_{AE} = 47.9 \text{ N } T$$

$$F_{DF} = -0.1815(47.9)$$

$$F_{DF} = 8.70 \text{ N } C$$

**13 Plane Motion of Rigid Bodies: Forces and Accelerations**

# Sample Problem 2

## ⚙️ Reflect and Think

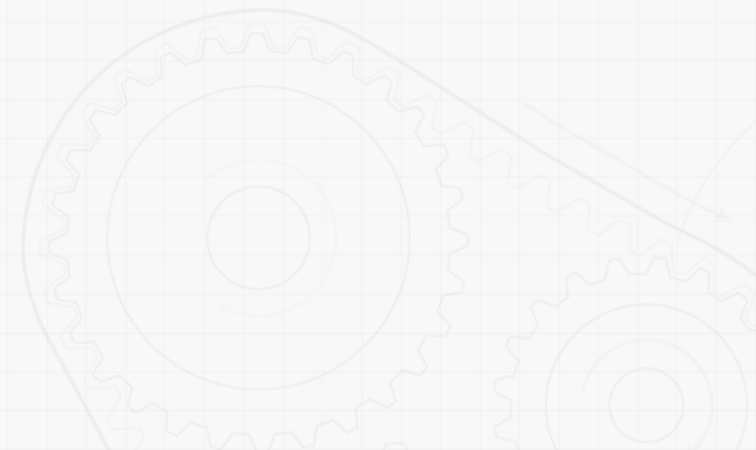
If AE and DF had been cables rather than links, the answers you just determined indicate that DF would have gone slack (that is you can't push on a rope), since the analysis showed that it would be in compression.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 2

### ⚙️ Reflect and Think

Therefore, the plate would not be undergoing curvilinear translation, but it would have been undergoing general plane motion.



**13** Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 2

## ⚙️ Reflect and Think

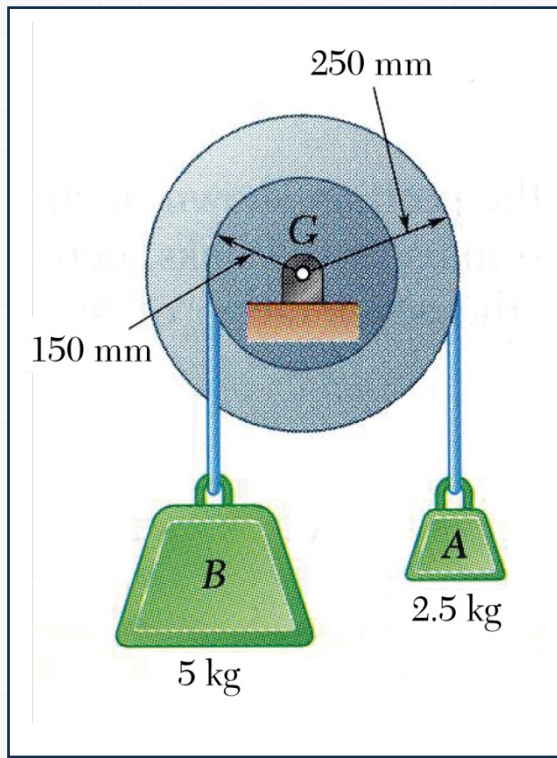
It is important to note that there is always more than one way to solve problems like this, since you can choose to take moments about any point you wish.

In this case, you took them about G, but you could have also chosen to take them about A or D.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

## Sample Problem 3

⚙️ Two pulleys are connected to the block.



- A pulley of mass **6kg** and having a radius of gyration of **200mm** is connected to two blocks as shown.
  - Assuming no axle friction, determine the angular acceleration of the pulley and the acceleration of each block.

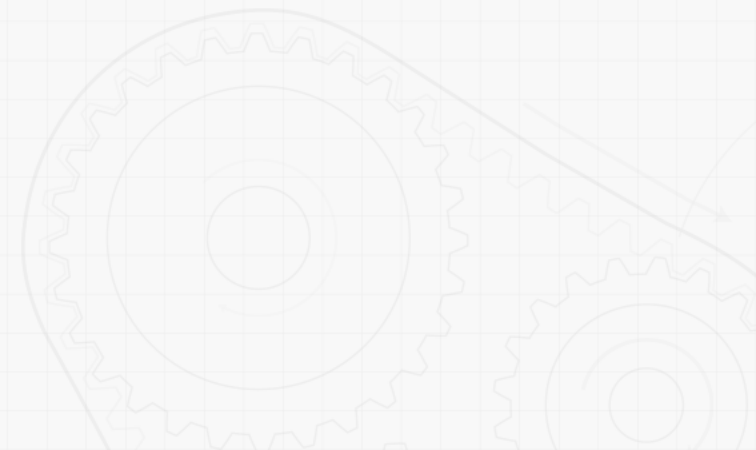
## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 3

⚙️ **Two pulleys are connected to the block.**

• **Strategy**

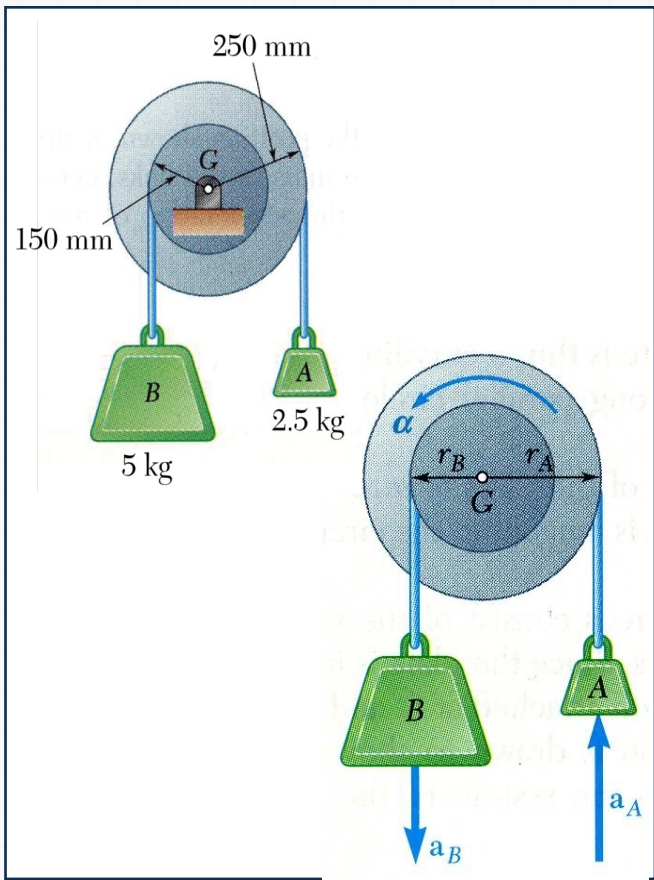
- Determine **the direction of rotation** by evaluating the net moment on the pulley due to the two blocks.
- Relate **the acceleration of the blocks** to **the angular acceleration of the pulley**.
- Draw **the free-body-diagram equation** expressing the equivalence of the external and effective forces on the complete pulley plus blocks system.
- Solve **the corresponding moment equation** for the pulley angular acceleration.



## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 3

## ⚙ Modeling and Analysis



### • Determine

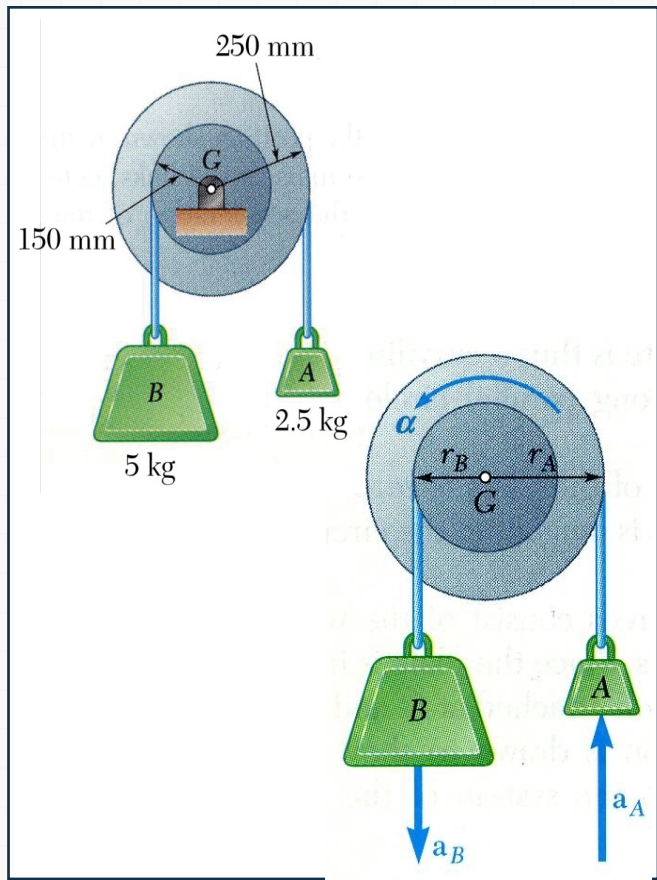
- The direction of rotation by evaluating the net moment on the pulley due to the two blocks.

$$+\circlearrowleft \sum M_G = (5)(9.81)(0.15) - (2.5)(9.81)(0.25) \\ = 1.22625 \text{ N} \cdot \text{m}$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 3

## Modeling and Analysis

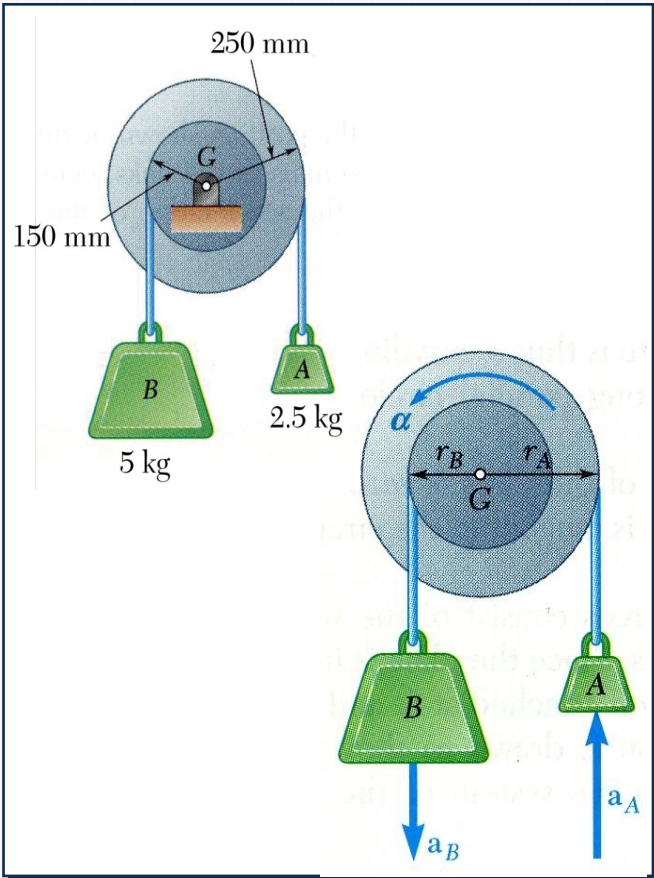


- Determine
  - Rotation is counterclockwise.
- Note

$$\bar{I} = m\bar{k}^2 = (6)(0.2)^2 = 0.24 \text{ kg} \cdot \text{m}^2$$

## Sample Problem 3

## ⚙️ Modeling and Analysis



- Relate the acceleration of the blocks to the angular acceleration of the pulley.

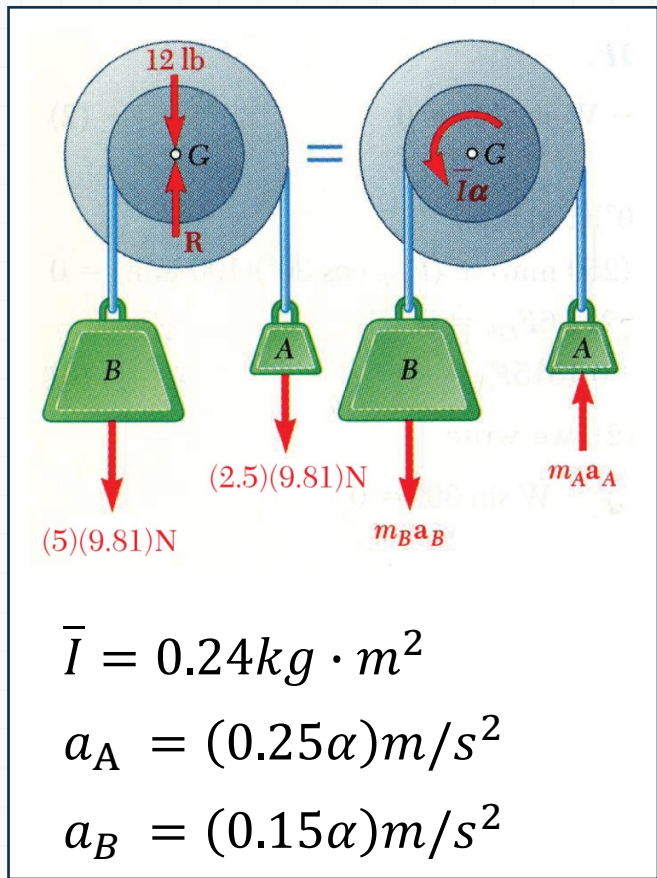
$$\begin{aligned} a_A &= r_A \alpha \\ &= (0.25) \alpha \end{aligned}$$

$$\begin{aligned} a_B &= r_B \alpha \\ &= (0.15) \alpha \end{aligned}$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 3

## Modeling and Analysis



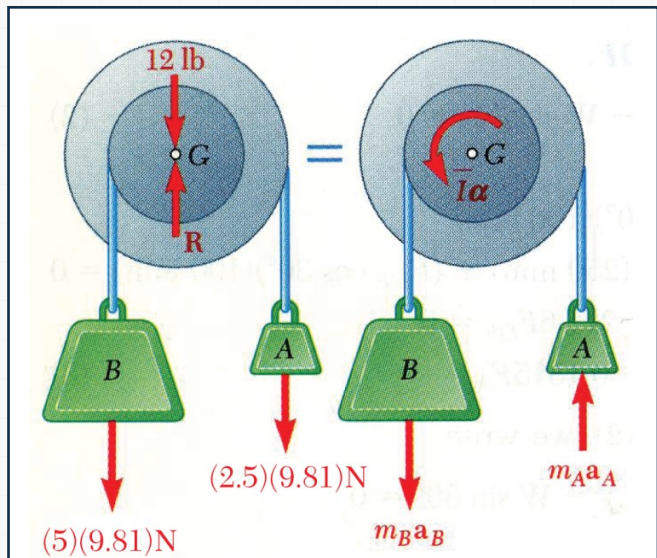
### Draw

- The free-body-diagram equation expressing the equivalence of the external
- Effective forces on the complete pulley and blocks system

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 3

## ⚙ Modeling and Analysis



$$\bar{I} = 0.24 \text{ kg} \cdot \text{m}^2$$

$$a_A = (0.25\alpha) \text{ m/s}^2$$

$$a_B = (0.15\alpha) \text{ m/s}^2$$

- Solve the corresponding moment equation for the pulley angular acceleration.

$$+\circlearrowleft \sum M_G = \sum (M_G)_{eff} \text{ (RHS from Kinetic Diagram)}$$

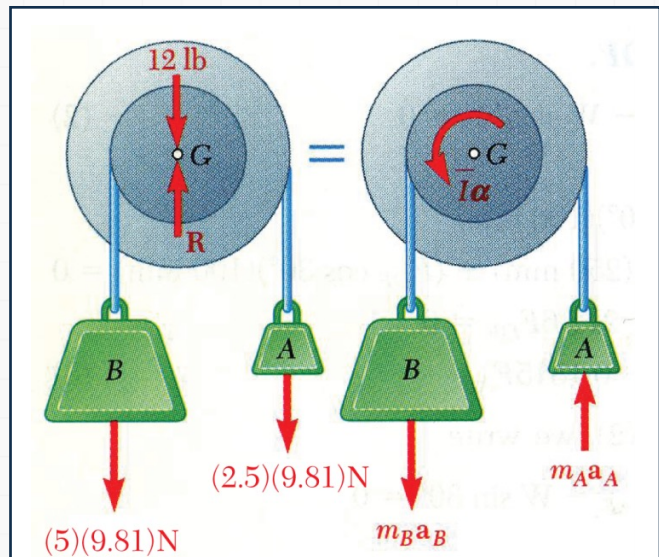
$$(5)(9.81)(0.15) - (2.5)(9.81)(0.25)$$

$$= \bar{I}\alpha + m_B a_B (0.15) - m_A a_A (0.25)$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 3

## Modeling and Analysis



$$\bar{I} = 0.24 \text{ kg} \cdot \text{m}^2$$

$$a_A = (0.25\alpha) \text{ m/s}^2$$

$$a_B = (0.15\alpha) \text{ m/s}^2$$

- Solve the corresponding moment equation for the pulley angular acceleration.

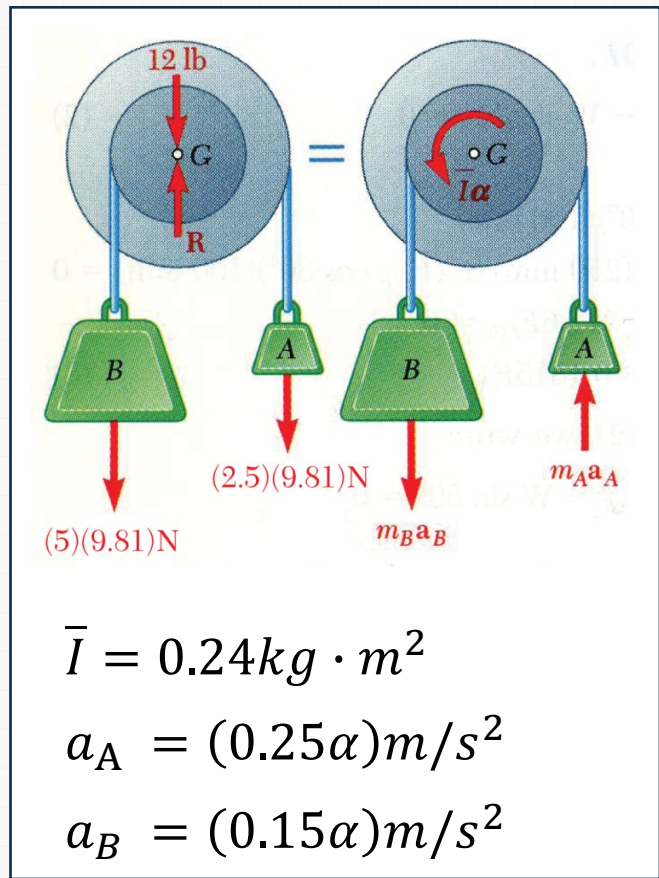
$$7.3575 - 6.1312 = (0.24)\alpha + 5(0.15\alpha)(0.15) - 2.5(0.25\alpha)(0.25)$$

$$\alpha = 2.41 \text{ rad/s}^2$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 3

## Modeling and Analysis



- Solve the corresponding moment equation for the pulley angular acceleration.

$$\begin{aligned} a_A &= r_A \alpha \\ &= (0.25)(2.41) \end{aligned}$$

$$a_A = 0.603 \text{ m/s}^2 \uparrow$$

$$\begin{aligned} a_B &= r_B \alpha \\ &= (0.15)(2.41) \end{aligned}$$

$$a_B = 0.362 \text{ m/s}^2 \downarrow$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

### Sample Problem 3

#### ⚙️ Reflect and Think

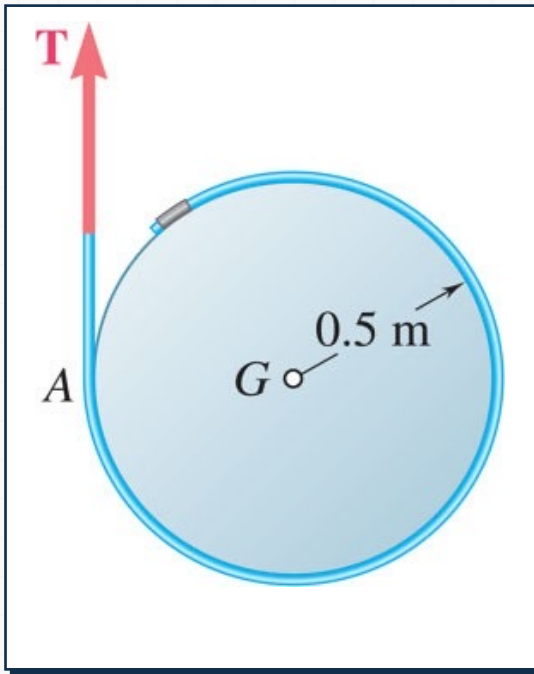
You could also solve this problem by considering the pulley and each block as separate systems, but you would have more resulting equations.

You would have to use this approach if you wanted to know the forces in the cables.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

## Sample Problem 4

⚙️ A cord is wrapped around a homogeneous disk of mass 15kg.



- The cord is pulled upwards with a force  $T = 180\text{ N}$ .
- Determine
  - The acceleration of the center of the disk
  - The angular acceleration of the disk
  - The acceleration of the cord

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

## Sample Problem 4

⚙️ **A cord is wrapped around a homogeneous disk of mass 15kg.**

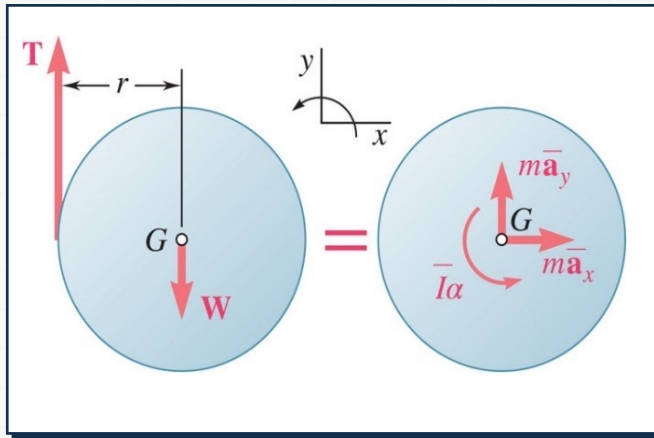
• **Strategy**

- Draw **the free-body-diagram equation** expressing the equivalence of the external and effective forces on the disk.
- Solve **the three corresponding scalar equilibrium equations** for the horizontal, vertical, and angular accelerations of the disk.
- Determine **the acceleration of the cord** by evaluating the tangential acceleration of the point A on the disk.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 4

## ⚙ Modeling and Analysis



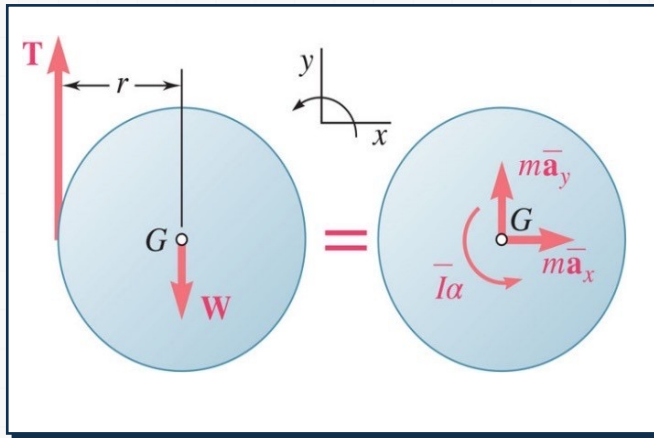
### • Draw

- The free-body-diagram equation expressing the equivalence of the external
- Effective forces on the disk.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 4

## ⚙ Modeling and Analysis



- Solve the three scalar equilibrium equations.

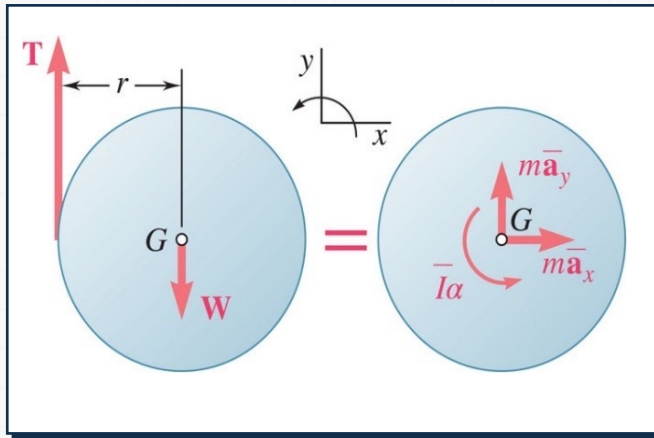
$$\sum^+ F_x = m\bar{a}_x = 0$$

$$\bar{a}_x = 0$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 4

## ⚙ Modeling and Analysis



- Solve the three scalar equilibrium equations.

$$+\uparrow \sum F_y = m\bar{a}_y$$

$$T - W = m\bar{a}_y$$

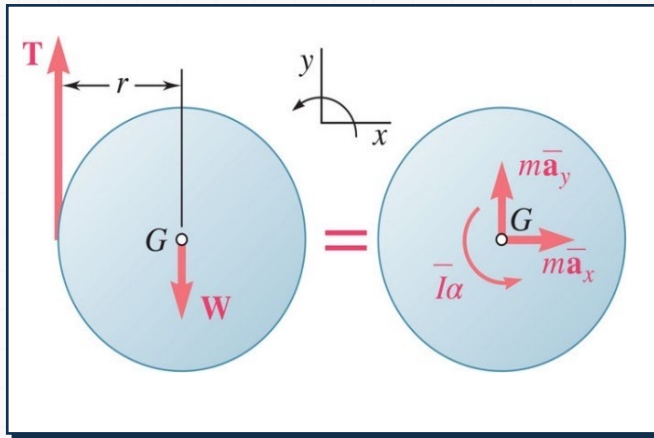
$$\bar{a}_y = \frac{T - W}{m} = \frac{180 - (15)(9.81)}{15}$$

$$\bar{a}_y = 2.19 \text{ m/s}^2 \uparrow$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 4

## ⚙ Modeling and Analysis



- Solve the three scalar equilibrium equations.

$$+\circlearrowleft \sum M_G = \bar{I}\alpha$$
$$-Tr = \bar{I}\alpha = \left(\frac{1}{2}mr^2\right)\alpha$$

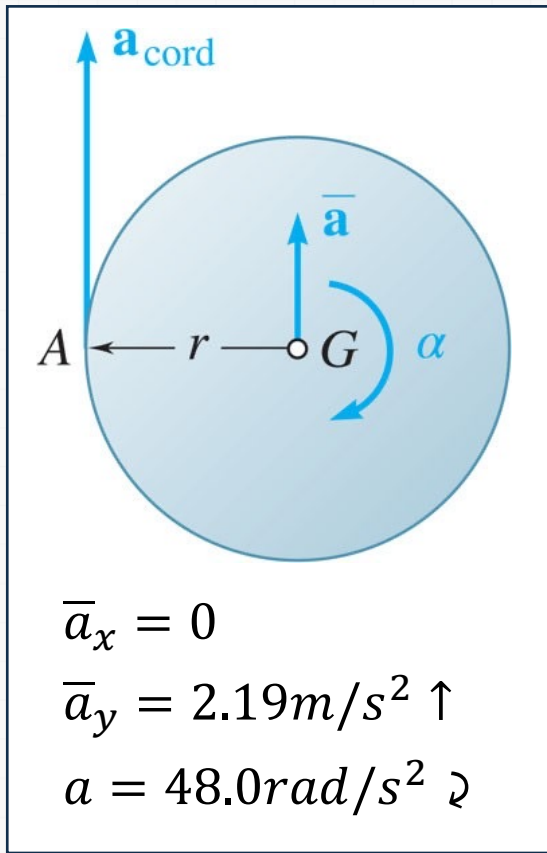
$$\alpha = -\frac{2T}{mr} = -\frac{2(180)}{15(0.5)}$$

$$\alpha = 48.0 \text{ rad/s}^2 \curvearrowright$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 4

## ⚙ Modeling and Analysis



### • Determine

- The acceleration of the cord by evaluating the tangential acceleration of the point A on the disk.

$$\begin{aligned}\vec{a}_{\text{cord}} &= (a_A)_t = \bar{a} + (a_{A/G})_t \\ &= 2.19 + (0.5)(48)\end{aligned}$$

$$a_{\text{cord}} = 26.2 \text{ m/s}^2 \uparrow$$

**13 Plane Motion of Rigid Bodies: Forces and Accelerations**

# Sample Problem 4

## ⚙️ Reflect and Think

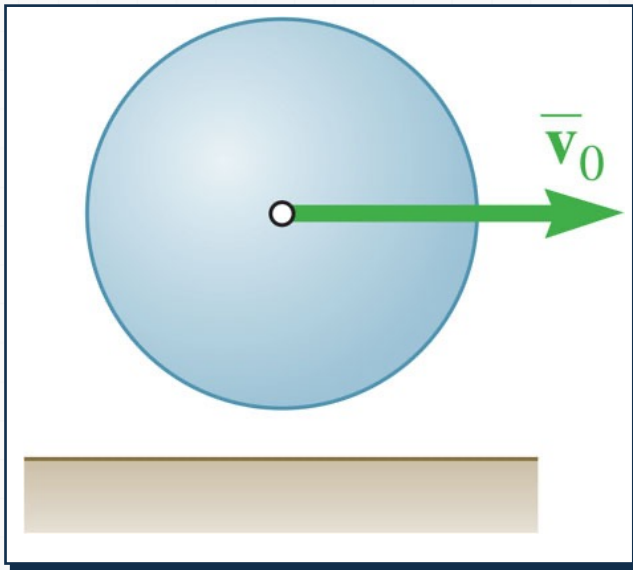
The angular acceleration is clockwise, as we would expect. A similar analysis would apply in many practical situations, such as pulling wire off a spool or paper off a roll.

In such cases, you would need to be sure that the tension pulling on the disk is not larger than the tensile strength of the material.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

## Sample Problem 5

⚙️ A uniform sphere of mass  $m$  and radius  $r$

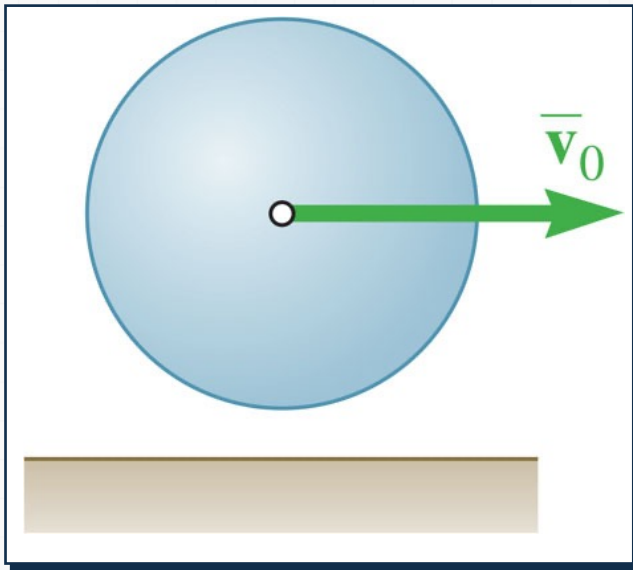


- A uniform sphere of mass  $m$  and radius  $r$  is projected along a rough horizontal surface with a linear velocity  $v_0$ .
- The coefficient of kinetic friction between the sphere and the surface is  $\mu_k$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

## Sample Problem 5

⚙️ A uniform sphere of mass  $m$  and radius  $r$



- Determine

- The time  $t_1$  at which the sphere will start rolling without sliding
- The linear and angular velocities of the sphere at time  $t_1$ .

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 5

⚙️ **A uniform sphere of mass  $m$  and radius  $r$**

• **Strategy**

- Draw **the free-body-diagram equation** expressing the equivalence of the external and effective forces on the sphere.
- Solve **the three corresponding scalar equilibrium equations** for the normal reaction from the surface and the linear and angular accelerations of the sphere.

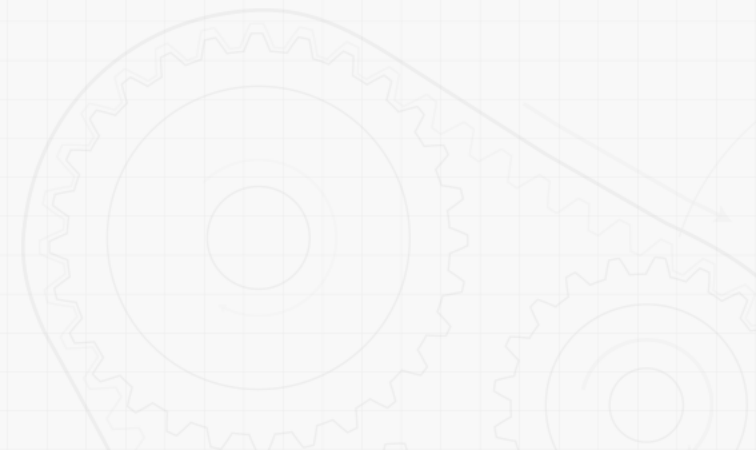
## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 5

⚙️ **A uniform sphere of mass  $m$  and radius  $r$**

• **Strategy**

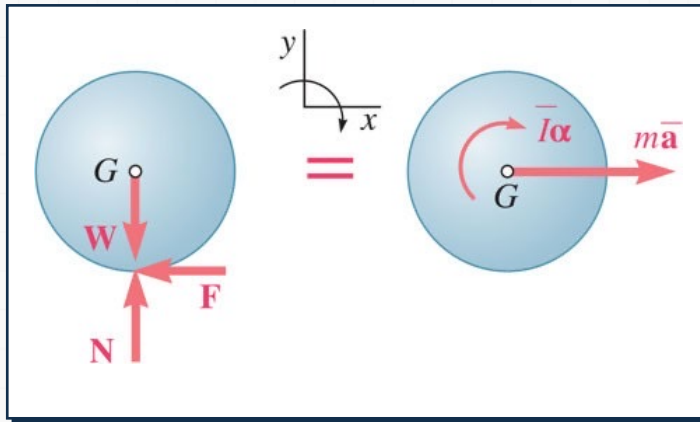
- Apply the kinematic relations for uniformly accelerated motion to determine the time at which the tangential velocity of the sphere at the surface is zero, that is **when the sphere stops sliding**.



## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 5

## ⚙ Modeling and Analysis



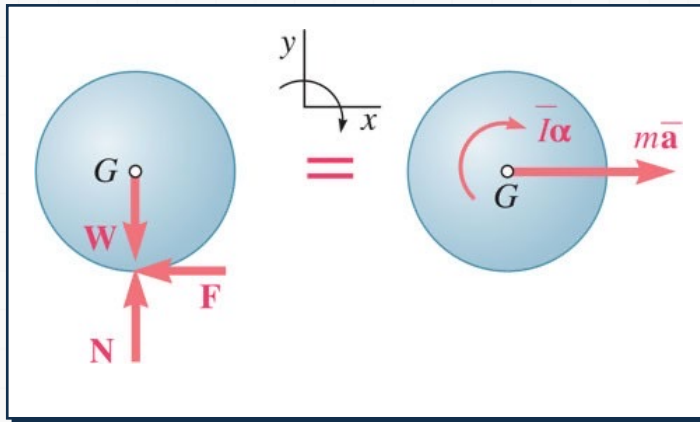
- Draw

- The free-body-diagram equation expressing the equivalence of the external
- Effective forces on the disk.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 5

## ⚙ Modeling and Analysis



- Solve the three scalar equilibrium equations.

$$+\uparrow \sum F_y = m\bar{a}_y$$

$$N - W = 0$$

$$+\rightarrow \sum F_x = \bar{a}_x$$

$$-F = m\bar{a}$$

$$-\mu_k mg = m\bar{a}$$

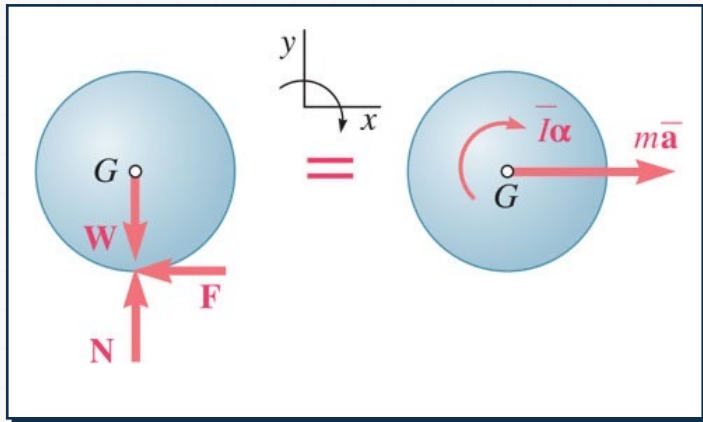
$$N = W = mg$$

$$\bar{a} = -\mu_k g$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 5

## ⚙ Modeling and Analysis



- Solve the three scalar equilibrium equations.

$$+\curvearrowright \sum M_G = \bar{I}\alpha$$

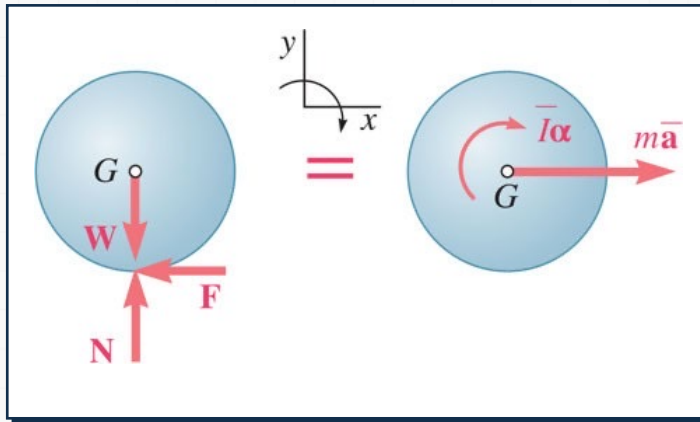
$$Fr = \bar{I}\alpha$$

$$(\mu_k mg)r = \left(\frac{2}{5}mr^2\right)\alpha \quad \alpha = \frac{5\mu_k g}{2r}$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 5

## ⚙ Modeling and Analysis



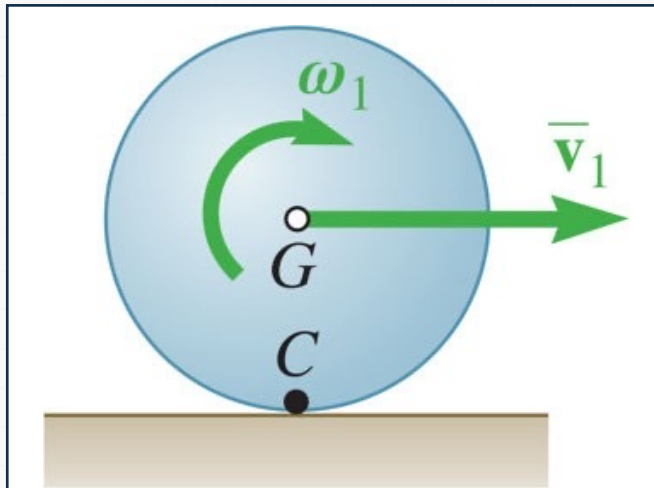
- Note

- As long as the sphere both rotates and slides, its linear
- Angular motions are uniformly accelerated.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 5

## ⚙ Modeling and Analysis



$$\bar{a} = -\mu_k g$$

$$\alpha = \frac{5}{2} \frac{\mu_k g}{r}$$

- Apply the kinematic relations for uniformly accelerated motion to determine the time at which the tangential velocity of the sphere at the surface is zero, that is when the sphere stops sliding.

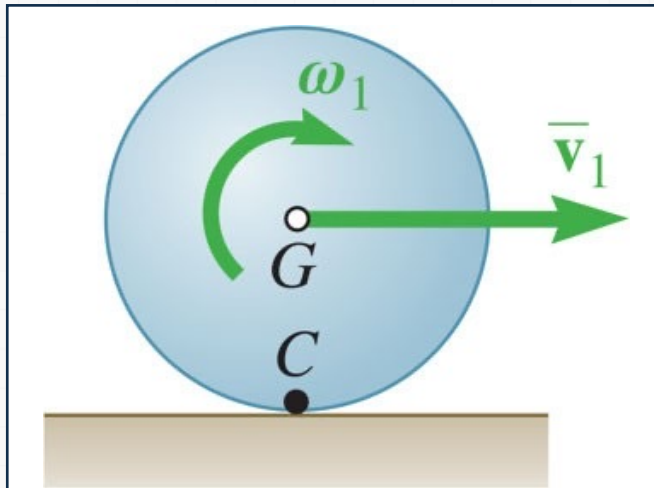
$$\bar{v} = \bar{v}_0 + \bar{a}t = \bar{v}_0 - \mu_k g t$$

$$\omega = \omega_0 + \alpha t = 0 + \left( \frac{5}{2} \frac{\mu_k g}{r} \right) t$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 5

## ⚙ Modeling and Analysis



$$\bar{a} = -\mu_k g$$

$$\alpha = \frac{5}{2} \frac{\mu_k g}{r}$$

- At the instant  $t_1$  when the sphere stops sliding,

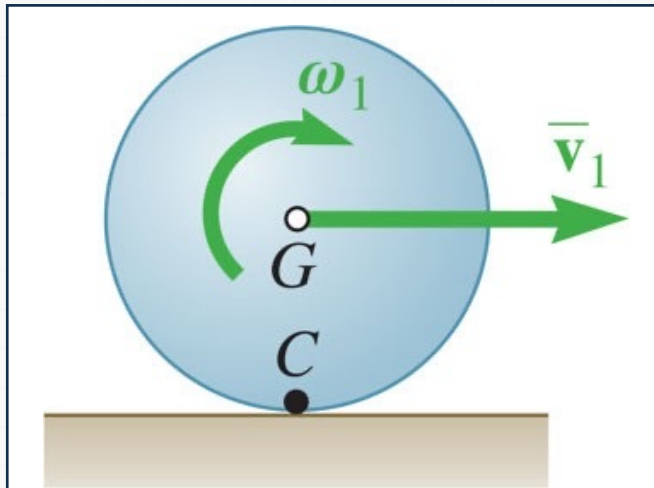
$$\bar{v}_1 = r\omega_1$$

$$\bar{v}_0 - \mu_k g t_1 = r \left( \frac{5}{2} \frac{\mu_k g}{r} \right) t_1 \quad t_1 = \frac{2}{7} \frac{\bar{v}_0}{\mu_k g}$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 5

## ⚙ Modeling and Analysis



$$\bar{a} = -\mu_k g$$

$$\alpha = \frac{5}{2} \frac{\mu_k g}{r}$$

- At the instant  $t_1$  when the sphere stops sliding,

$$\omega_1 = \left( \frac{5}{2} \frac{\mu_k g}{r} \right) t_1 = \left( \frac{5}{2} \frac{\mu_k g}{r} \right) \left( \frac{2}{7} \frac{\bar{v}_0}{\mu_k g} \right)$$

$$\omega_1 = \frac{5}{7} \frac{\bar{v}_0}{r}$$

$$\bar{v}_1 = r \omega_1 = r \left( \frac{5}{7} \frac{\bar{v}_0}{r} \right)$$

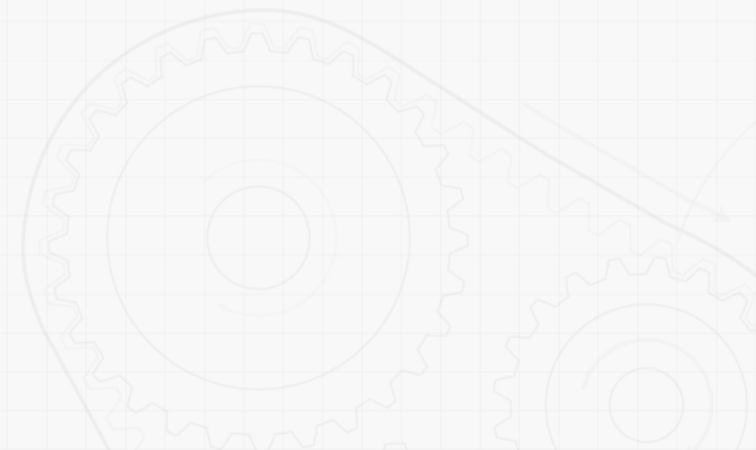
$$\bar{v}_1 = \frac{5}{7} \bar{v}_0$$

### 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 5

## ⚙️ Reflect and Think

You could use this type of analysis to determine how long it takes a bowling ball to begin to roll without slip or to see how the coefficient of friction affects this motion.



## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 5

## ⚙️ Reflect and Think

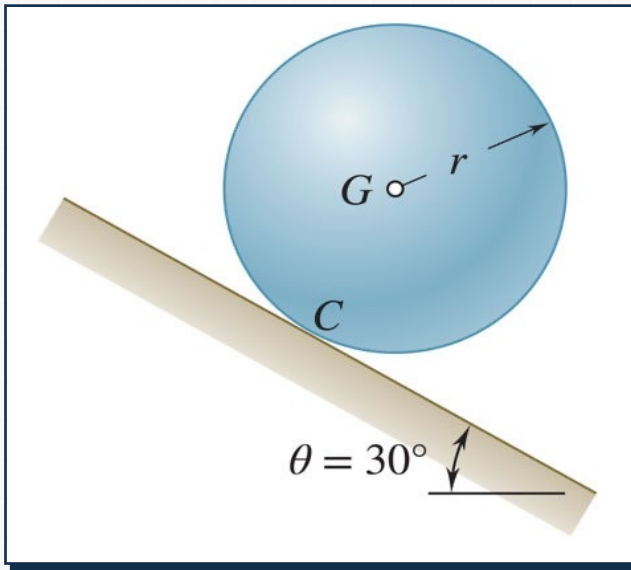
Instead of taking moments about the center of gravity, you could have chosen to take moments about point C, in which case your third equation would have been

$$\sum M_c = H_c \rightarrow 0 = m\bar{a}r + I\bar{\alpha}$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

## Sample Problem 6

⚙️ A sphere of weight  $W$



- It is released with no initial velocity and rolls without slipping on the incline.
- Determine
  - The minimum value of the coefficient of friction
  - The velocity of  $G$  after the sphere has rolled 10ft
  - The velocity of  $G$  if the sphere were to move 10ft down a frictionless incline.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 6

## ⚙️ A sphere of weight $W$

### • Strategy

- Draw the free-body-equation for the sphere, expressing the equivalence of the external and effective forces.
- With the linear and angular accelerations related, solve the three scalar equations derived from the free-body-equation for the angular acceleration and the normal and tangential reactions at C.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 6

## ⚙️ A sphere of weight $W$

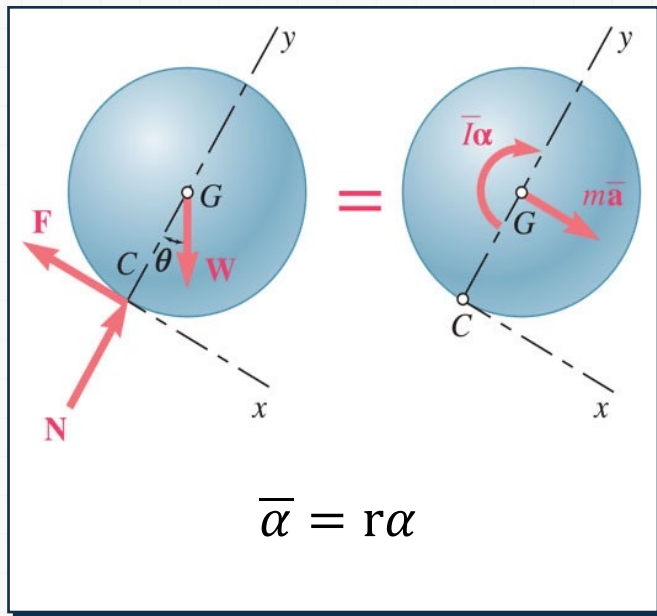
### • Strategy

- Calculate the friction coefficient required for the indicated tangential reaction at C.
- Calculate the velocity after 10ft of uniformly accelerated motion.
- Assuming no friction, calculate the linear acceleration down the incline and the corresponding velocity after 10ft.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 6

## ⚙ Modeling and Analysis



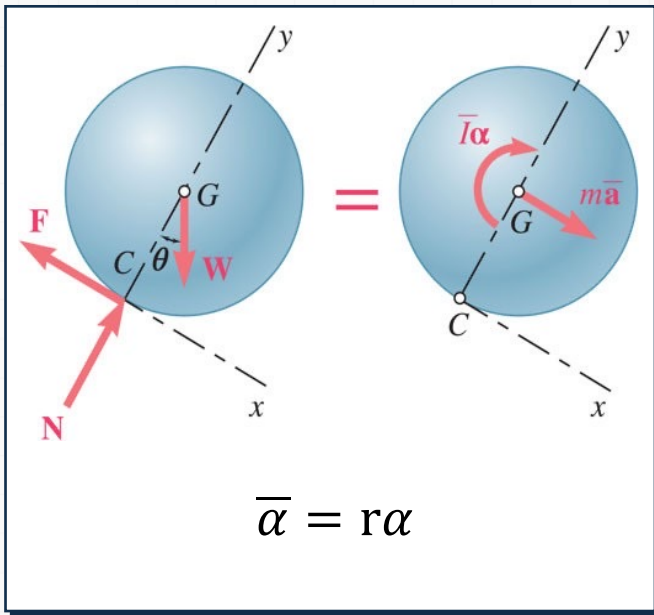
### • Draw

- The free-body-equation for the sphere, expressing the equivalence of the external
- Effective forces

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 6

## ⚙ Modeling and Analysis

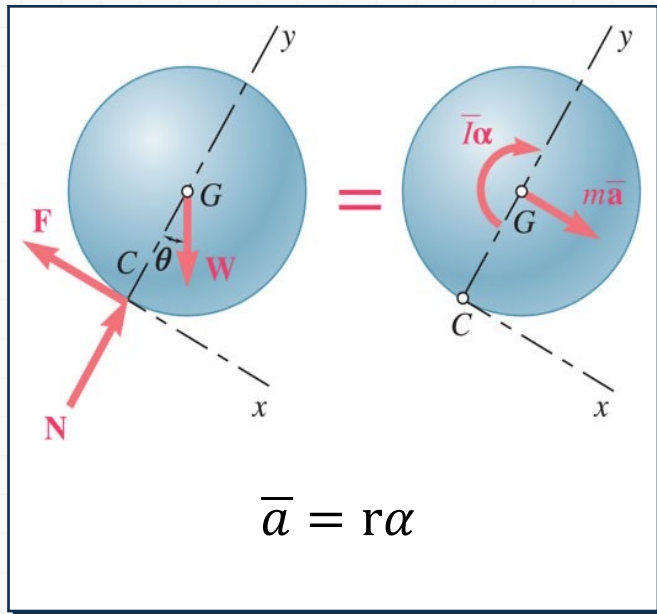


- With the linear and angular accelerations related, solve **the three scalar equations** derived from the free-body-equation for the angular acceleration and the normal and tangential reactions at C.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 6

## Modeling and Analysis



$$\sum M_C = \bar{I}\alpha + m\bar{a}d_{\perp}$$

$$(W \sin \theta)r = m\bar{a}r + \bar{I}\alpha$$

$$= mr\alpha r + \frac{2}{5}mr^2\alpha$$

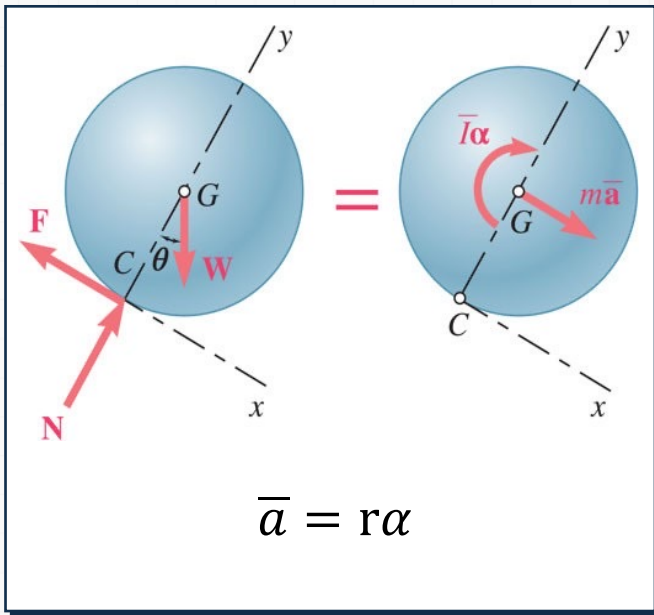
$$= \frac{W}{g}r^2\alpha + \frac{2}{5}\frac{W}{g}r^2\alpha$$

$$\alpha = \frac{5g\sin\theta}{7r}$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 6

## ⚙ Modeling and Analysis



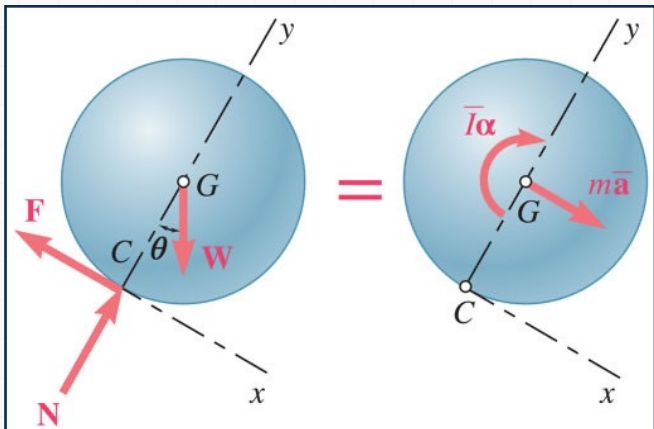
$$\begin{aligned}\bar{a} &= r\alpha = \frac{5g \sin 30^\circ}{7} \\ &= \frac{5(9.81) \sin 30^\circ}{7}\end{aligned}$$

$$\bar{a} = 3.50 \text{ m/s}^2$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 6

## ⚙ Modeling and Analysis



$$\alpha = \frac{5g\sin\theta}{7r}$$

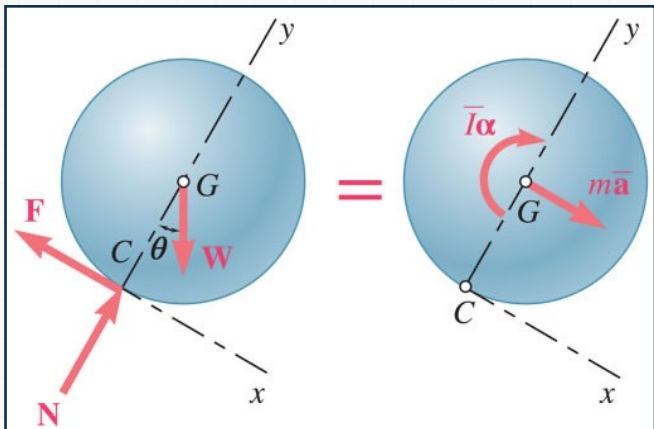
$$\bar{a} = r\alpha = 11.50\text{ft/s}^2$$

- Solve **the three scalar equations** derived from the free-body-equation for the angular acceleration and **the normal and tangential reactions at C**

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 6

## Modeling and Analysis



$$\alpha = \frac{5g \sin \theta}{7r}$$

$$\bar{a} = r\alpha = 11.50 \text{ ft/s}^2$$

$$\sum F_x = m\bar{a}_x \quad W \sin \theta - F = m\bar{a}$$

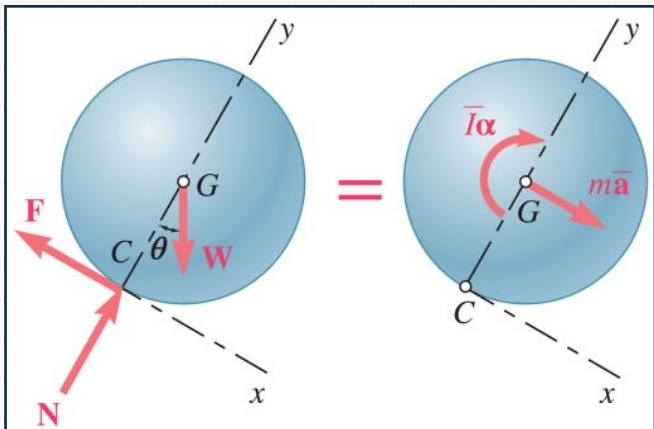
$$= \frac{W}{g} \frac{5g \sin \theta}{7}$$

$$F = \frac{2}{7} W \sin 30^\circ = 0.143W$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 6

## Modeling and Analysis



$$\alpha = \frac{5g\sin\theta}{7r}$$

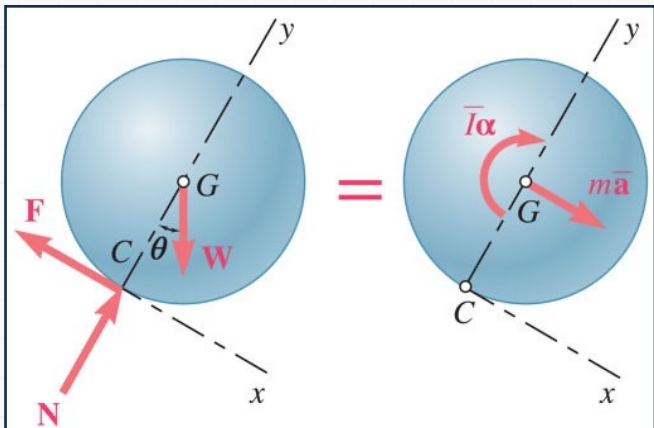
$$\bar{a} = r\alpha = 11.50\text{ft/s}^2$$

$$\sum F_y = m\bar{a}_y \quad N - W\cos\theta = 0$$
$$N = W\cos 30^\circ = 0.866W$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 6

## ⚙ Modeling and Analysis



$$\alpha = \frac{5g \sin \theta}{7r}$$

$$\bar{a} = r\alpha = 11.50 \text{ ft/s}^2$$

- Calculate the friction coefficient required for the indicated tangential reaction at C.

$$F = \mu_s N$$

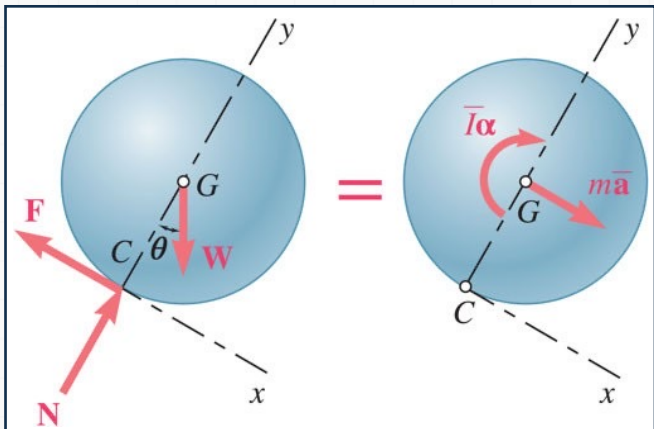
$$\mu_s = \frac{F}{N} = \frac{0.143W}{0.866W}$$

$$\mu_s = 0.165$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 6

## Modeling and Analysis



$$\alpha = \frac{5g \sin \theta}{7r}$$

$$\bar{a} = r\alpha = 3.504 \text{ m/s}^2$$

- Calculate the velocity after 10ft of uniformly accelerated motion.

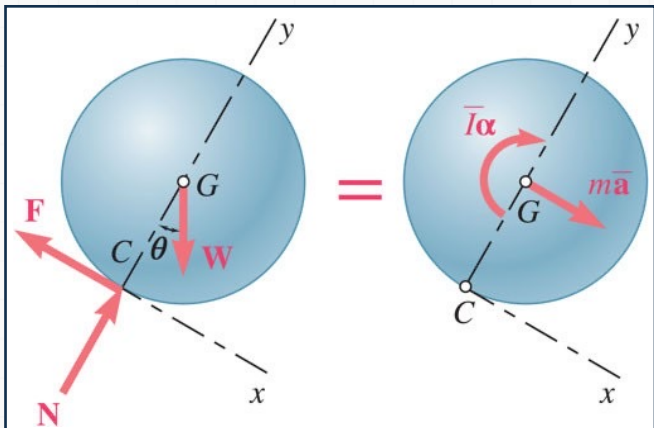
$$\begin{aligned} \bar{v}^2 &= \bar{v}_0^2 + 2\bar{a}(\bar{x} - \bar{x}_0) \\ &= 0 + 2(3.504)(3) \end{aligned}$$

$$v = 4.59 \text{ m/s}$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 6

## Modeling and Analysis



$$\alpha = \frac{5g \sin \theta}{7r}$$

$$\bar{a} = r\alpha = 3.504 \text{ m/s}^2$$

- Assuming no friction, calculate the linear acceleration and the corresponding velocity after 3m.

$$\sum M_G = \bar{I}\alpha$$

$$0 = \bar{I}\alpha \quad \alpha = 0$$

$$\sum F_x = m\bar{a}_x$$

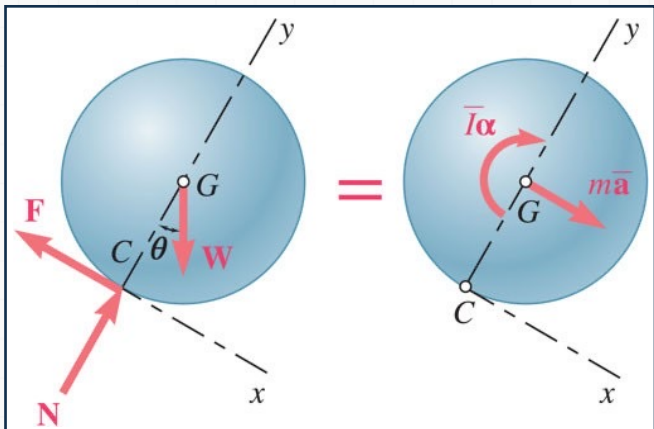
$$W \sin 30^\circ = mg \sin 30^\circ = m\bar{a}$$

$$\bar{a} = (9.81) \sin 30^\circ = 4.905 \text{ m/s}^2$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 6

## ⚙ Modeling and Analysis



$$\alpha = \frac{5g \sin \theta}{7r}$$

$$\bar{a} = r\alpha = 3.504 \text{ m/s}^2$$

- Assuming no friction, calculate the linear acceleration and the corresponding velocity after 3m.

$$\begin{aligned}\bar{v}^2 &= \bar{v}_0^2 + 2\bar{a}(\bar{x} - \bar{x}_0) \\ &= 0 + 2(4.905)(3)\end{aligned}$$

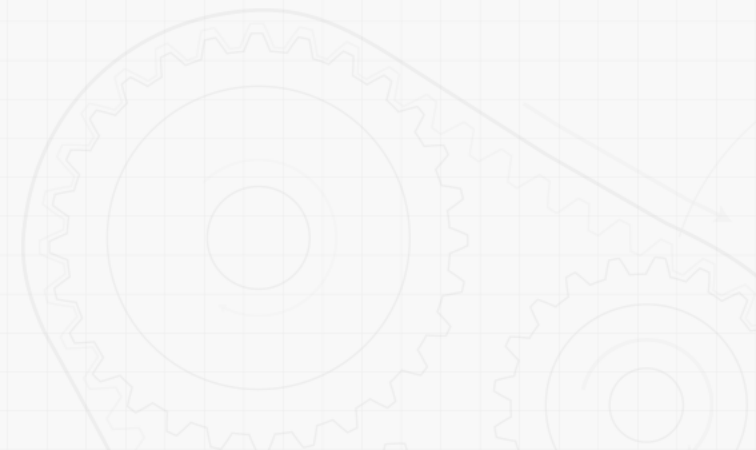
$$v = 5.42 \text{ m/s}$$

### 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 6

## ⚙️ Reflect and Think

Note that the sphere moving down a frictionless surface has a higher velocity than the rolling sphere, as you would expect.



## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 6

## ⚙️ Reflect and Think

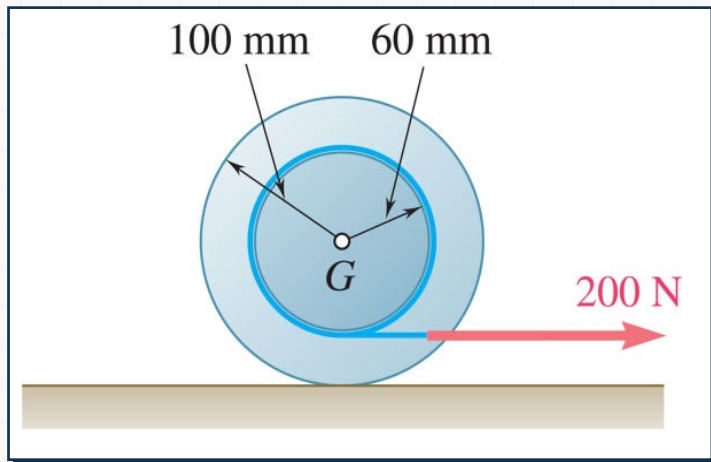
It is also interesting to note that the expression you obtained for the acceleration of the center of mass, that is,  $a = 5/7 g \sin \theta$ , is independent of the radius of the sphere and the mass of the sphere.

This means that any two solid spheres, as long they are rolling without sliding, have the same linear acceleration.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

## Sample Problem 7

### ⚙️ A cord wrapping the hub inside the wheel



- A cord is wrapped around the inner hub of a wheel and pulled horizontally with a force of **200 N**.
- The wheel has a mass of **50 kg** and a radius of gyration of **70 mm**
- Knowing  $\mu_s = 0.20$  and  $\mu_k = 0.15$ , determine the acceleration of G and the angular acceleration of the wheel.

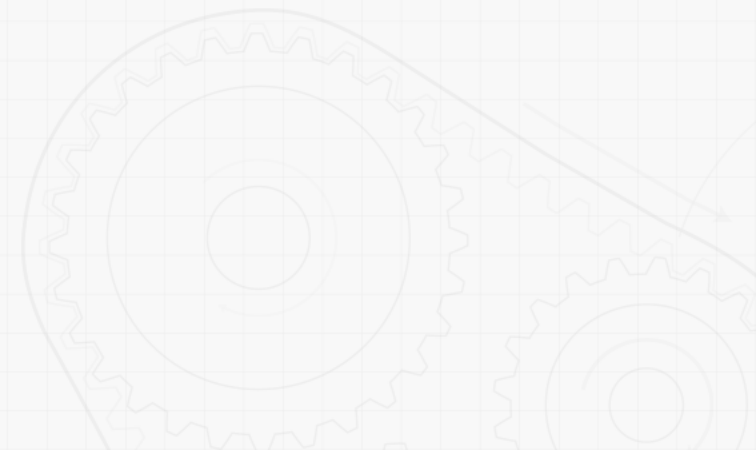
## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 7

## ⚙️ A cord wrapping the hub inside the wheel

### • Strategy

- Draw the free-body-equation for the wheel, expressing the equivalence of the external and effective forces.
- Assuming rolling without slipping and therefore, related linear and angular accelerations, solve the scalar equations for the acceleration and the normal and tangential reactions at the ground.



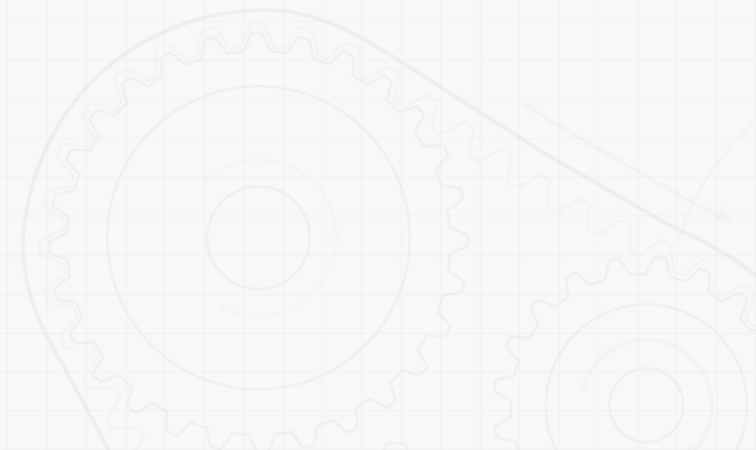
## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 7

## ⚙️ A cord wrapping the hub inside the wheel

### • Strategy

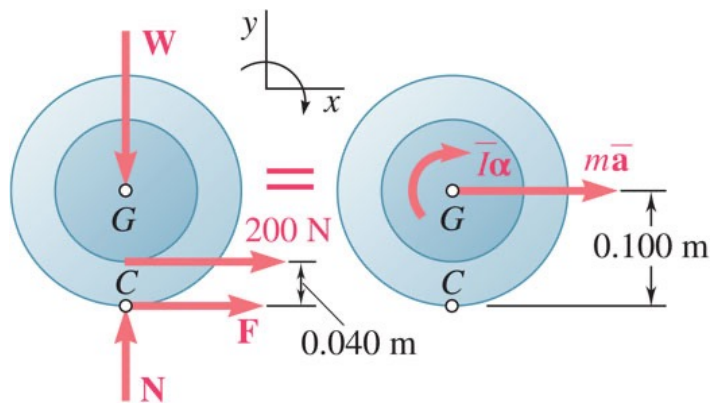
- Compare the required tangential reaction to the maximum possible friction force.
- If slipping occurs, calculate the kinetic friction force and then solve the scalar equations for the linear and angular accelerations.



## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 7

## ⚙️ Modeling and Analysis



- Draw
  - The free-body-equation for the wheel
- Assuming rolling without slipping, solve **the scalar equations** for the acceleration and ground reactions.

$$\begin{aligned}\bar{I} &= m\bar{k}^2 = (50)(0.70)^2 \\ &= 0.245 \text{ kg} \cdot \text{m}^2\end{aligned}$$

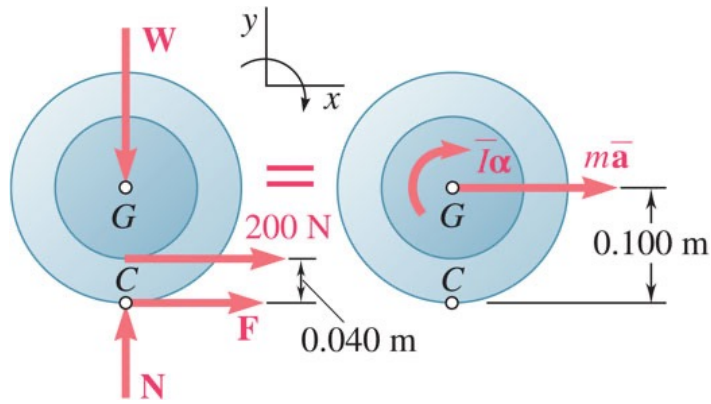
Assume rolling without slipping,

$$\bar{a} = r\alpha = (0.100)\alpha$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 7

## Modeling and Analysis



$$\begin{aligned}\bar{I} &= m\bar{k}^2 = (50)(0.70)^2 \\ &= 0.245 \text{ kg} \cdot \text{m}^2\end{aligned}$$

Assume rolling without slipping,

$$\bar{a} = r\alpha = (0.100)\alpha$$

$$+\circlearrowleft \sum M_c = \bar{I}\alpha + m\bar{a}d_{\perp}$$

$$(200)(0.040) = m\bar{a}(0.100) + \bar{I}\alpha$$

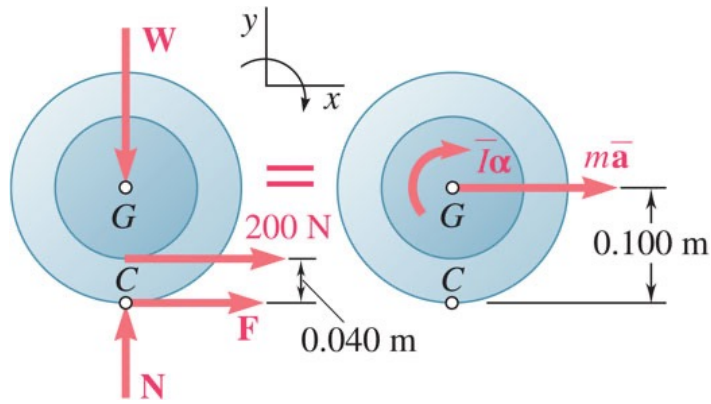
$$8.0 = (50)(0.100)^2\alpha + (0.245)\alpha$$

$$\alpha = +10.74 \text{ rad/s}^2$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 7

## ⚙ Modeling and Analysis



$$\begin{aligned}\bar{I} &= m\bar{k}^2 = (50)(0.070)^2 \\ &= 0.245 \text{ kg} \cdot \text{m}^2\end{aligned}$$

Assume rolling without slipping,

$$\bar{a} = r\alpha = (0.100)\alpha$$

$$\bar{a} = (0.100)(10.74) = 1.074 \text{ m/s}^2 \rightarrow$$

$$\rightarrow \sum F_x = m\bar{a}_x$$

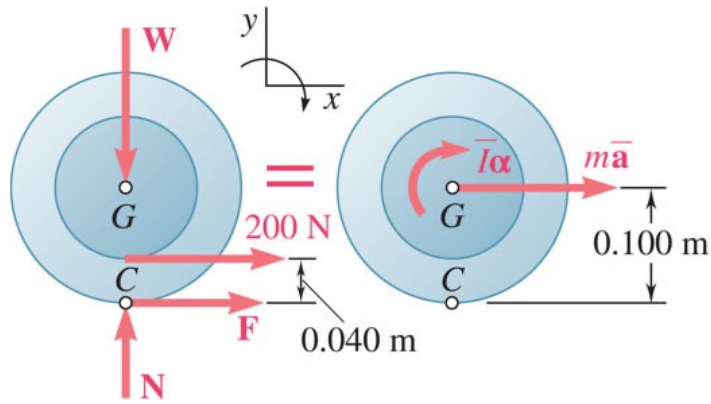
$$F + 200 \text{ N} = m\bar{a} = (50)(1.074)$$

$$F = -146.3 \text{ N} (\leftarrow)$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 7

## ⚙ Modeling and Analysis



$$\begin{aligned}\bar{I} &= m\bar{k}^2 = (50)(0.70)^2 \\ &= 0.245 \text{ kg} \cdot \text{m}^2\end{aligned}$$

Assume rolling without slipping,

$$\bar{a} = r\alpha = (0.100)\alpha$$

$$+\uparrow \sum F_y = m\bar{a}_y$$

$$N - W = 0$$

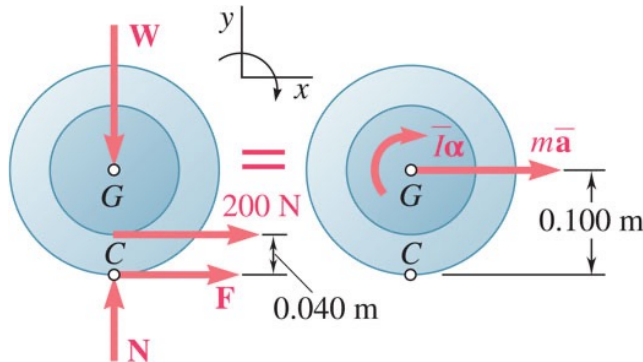
$$N = mg = (50)(9.81) = 490.5 \text{ N } (\uparrow)$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 7

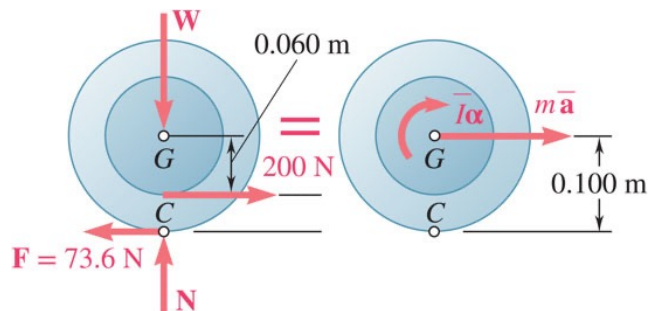
## Modeling and Analysis

- Compare the required tangential reaction to the maximum possible friction force.



Without slipping

$$F = -146.3\text{ N}, N = 490.5\text{ N}$$



$$F_{max} = \mu_s N = 0.20(490.5\text{ N}) = 98.1\text{ N}$$

$$F > F_{max},$$

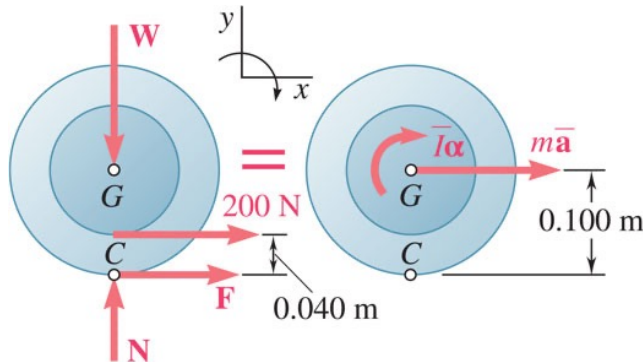
rolling without slipping is impossible.

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 7

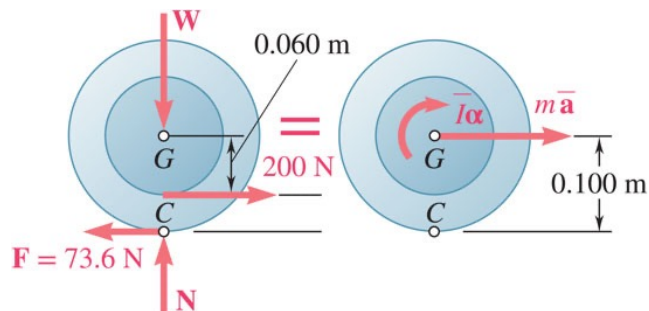
## Modeling and Analysis

- Calculate the friction force with slipping and solve the scalar equations for linear and angular accelerations.



Without slipping

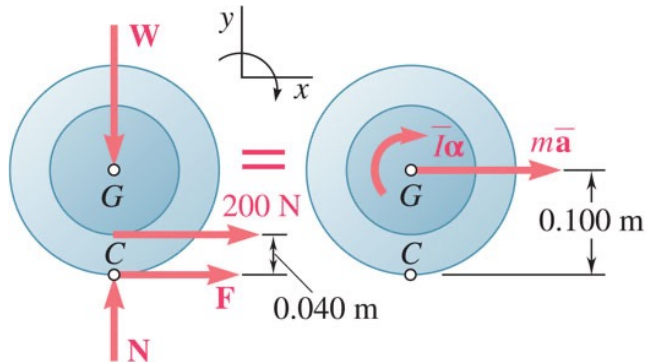
$$F = -146.3\text{ N}, N = 490.5\text{ N}$$



## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

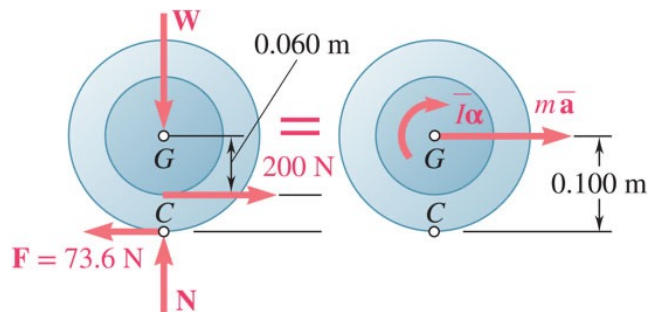
## Sample Problem 7

## Modeling and Analysis



Without slipping

$$F = -146.3\text{ N}, N = 490.5\text{ N}$$



$$F = F_k = \mu_k N = 0.15(490.5) = 73.6\text{ N}$$

$$\sum F_x = m\bar{a}_x$$

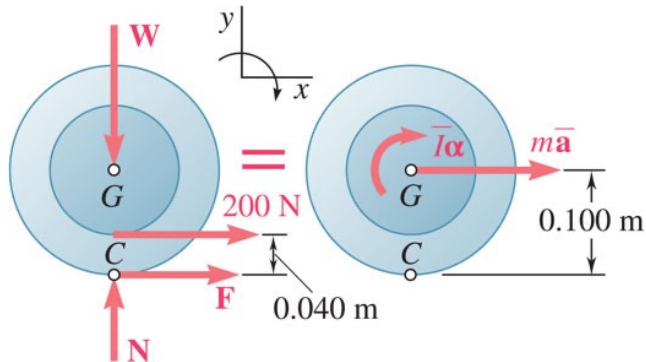
$$200 - 73.6 = (50)\bar{a}$$

$$\bar{a} = 2.53\text{ m/s}^2$$

## 13 Plane Motion of Rigid Bodies: Forces and Accelerations

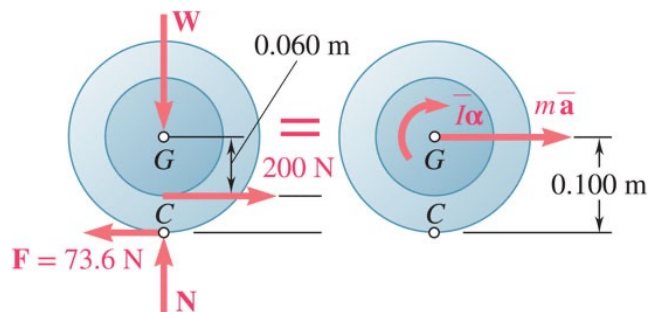
## Sample Problem 7

## Modeling and Analysis



Without slipping

$$F = -146.3\text{ N}, N = 490.5\text{ N}$$



$$+\curvearrowright \sum M_G = \bar{I}\alpha$$

$$(73.6)(0.100) - (200)(0.060)$$

$$= (0.245)\alpha$$

$$\alpha = -18.94\text{ rad/s}^2$$

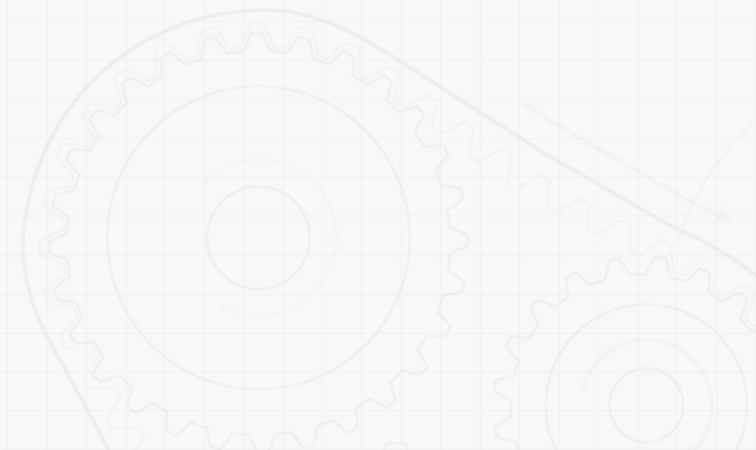
$$\alpha = 18.94\text{ rad/s}^2$$

### 13 Plane Motion of Rigid Bodies: Forces and Accelerations

# Sample Problem 7

## ⚙️ Reflect and Think

The wheel has larger linear and angular accelerations under conditions of rotating while sliding than when rolling without sliding.



### 13 Plane Motion of Rigid Bodies: Forces and Accelerations

## References

- Beer, Johnston, Cornwell, Self, Sanghi, "Vector Mechanics for Engineers: Dynamics" 12th Edition, McGraw Hill, Chapter 16